

Mapping multi-layered mangroves from multispectral, hyperspectral, and LiDAR data

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ABSTRACT

Understanding species distribution and canopy structure of mangrove forests is imperative for flora and fauna conservation in mangrove habitats. However, most mangrove studies focused on the top canopy layer without exploring the vertical structure of mangroves. This paper presents multi-layered mangrove mapping which considered both overstory and understory detection and species classification using multispectral WorldView-3 (WV-3) data, airborne hyperspectral images (HSI), and LiDAR point cloud. First, LiDAR returns were stratified into the overstory and understory by analyzing the profile of return height, which helped understand the vertical structure of the mangrove stands. Second, three classification algorithms Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Network (CNN) were compared by applying WV-3, HSI, LiDAR data, and their combinations to map seven vegetative species. Feature selection was conducted to identify important features and the optimal feature size prior to classification tasks. The measured and estimated understory canopy heights reached a high correlation coefficient of 0.71, which demonstrated the effectiveness of using LiDAR data and the proposed procedure to stratify multi-layered canopies. The combined HSI and LiDAR data produced satisfactory results by the three classifiers with overall accuracy (OA) varying from 0.86 to 0.88. And the species was also accurately mapped by integrating WV-3 and LiDAR data using both RF and SVM algorithms with OA attaining between 0.84 and 0.86. The results of this study highlight that (1) LiDAR data provided superior information to map the vertical structure of multi-layered mangroves, which provided valuable information to classify single-layered and dual-layered *Kandelia obovata* with understory beneath; (2) the combination of spectral and LiDAR features improved mangrove species classification; (3) and species mapping results derived from combined datasets appeared to be more influential by LiDAR features when using RF and SVM, but spectral features played a more important role in CNN.

1. Introduction

Mangrove communities or mangals are unique inter-tidal wetland ecosystems in the tropical and subtropical coastal areas (FAO, 2007), which have significant ecological value to the natural environment and economic value to human society. The mangal community maintains biodiversity and provides habitats for diverse aquatic plants, animals and attracts thousands of water birds. Besides, the mangrove ecosystem is a valuable resource for production, recreation, education, and scientific research (AFCD, 2019; Tam and Wong, 2000). However, mangroves face the loss of area, infection, or species invasion due to excessive agricultural activities, urbanization, natural hazard, or improper management globally. Therefore, understanding mangrove species

distribution and monitoring their growth are necessary to support the formulation of mangrove conservation measures. While the swampy environment and inaccessibility to mangrove forests often hinder field investigation, remote sensing technology has been applied in mangrove studies for the past three decades (Blasco et al., 1992; Chun et al., 2015; Green et al., 1998; Wang et al., 2004b, 2015), which is still a timely and efficient tool for mangrove mapping and monitoring.

The advent of high spatial resolution sensors such as SPOT (HVR, HRVIR, or HRG), Quickbird, IKONOS, and GeoEye that provide meter-level resolution images improved vegetation classification at the species level (Giri et al., 2014; Ju et al., 2003; Wang et al., 2004b, 2004a; Zhou et al., 2009). New generation sensors provide images with both high spatial resolution and novel bands at some spectral regions such as

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the red edge band which was found strongly related to chlorophyll concentration in leaves (Clevers, 1999; Mutanga and Skidmore, 2007). Sub-meter WorldView-2/3 (WV-2/3) images capturing the reflectance at the red edge, yellow, and NIR regions are important for diverse vegetation species classification such as mangrove stand, subtropical forest, coastal wetland, urban area, etc., and quite a number of studies demonstrated the power of the red edge band in the species discrimination (Rapinel et al., 2014; Sothe et al., 2018; Wang et al., 2015; Waser et al., 2014).

Hyperspectral data generally provides hundreds of contiguous spectral bands between 400 nm and 2500 nm, and with such spectral detail, it allows in-depth vegetation research. With the development of airborne sensors, high-resolution hyperspectral images attract much attention in the field of vegetation discrimination. These studies involved a wide range of forest types with various classification algorithms, which indicated the applicability and potential of hyperspectral images (Burai et al., 2015; Kamal and Phinn, 2011; Peerbhay et al., 2013; Richter et al., 2016). However, the extra spectral resolution also causes the Hughes phenomenon (Hughes, 1968) which occurs when an excessive number of features are applied to classify a certain number of training samples, but results in a decrease of classification accuracy. Previous studies suggested that feature selection could efficiently avoid the Hughes phenomenon and over-fitting problems in classification (Cheng et al., 2006; Guyon and Elisseeff, 2003). In some studies, the classification accuracy obtained from selected bands were reported to be even higher than those with full bands as input (Abdel-Rahman et al., 2014; Fassnacht et al., 2014; Li et al., 2019). The optimal subset size was observed to have a tendency from 15 to 20 bands/features for the best classification accuracy (Fassnacht et al., 2014). Besides, important features identified from feature selection results can shed some light on factors that affect the classification accuracy. According to previous studies, important spectral bands frequently retained the green peak, yellow edge, red valley, and red edge regions (450–600 nm, 680–750 nm), NIR region (800–900 nm), and SWIR region (1150–1250 nm, 1450–1500 nm, and 1950–2050 nm) situated close to water absorption regions (Fassnacht et al., 2014; Ghosh and Joshi, 2014; Shen and Cao, 2017; Treitz et al., 2010). Even though optical images were widely used for vegetation species identification, limitations of multispectral and hyperspectral data were also investigated. For instance, different tree species share similar spectral signatures, while the spectral signature of the same species may vary across the forest area due to environmental factors (Ghiyamat and Shafri, 2010; Immitzer et al., 2012).

Active light detection and ranging (LiDAR) data provides three-dimensional (3D) information which is attractive for vegetation studies. Former studies suggested that discrete LiDAR data with high point density was able to extract more detailed structural features such as distribution patterns of foliage and branch, which can improve species-level identification (Shi et al., 2018; Zhao and Popescu, 2009). Over the last decade, many geometric and radiometric LiDAR metrics were proposed to describe the characteristic of trees (Koenig and Höfle, 2016; Shi et al., 2018). Geometric metrics intuitively described the vertical structure and morphology while radiometric metrics derived from intensity signal to reflect foliage type, leaf size, leaf orientation, leaf clumping, and foliage density (Koenig and Höfle, 2016; Zhao and Popescu, 2009). Previous studies documented the successes of either leaf-on and leaf-off LiDAR data to discriminate species for deciduous forests, coniferous forests, and urban trees (Hamraz et al., 2017; Shi et al., 2018; Singh et al., 2015; Teobaldelli et al., 2017). Furthermore, the synergy of the optical image and LiDAR point clouds can significantly improve the vegetation species classification. For example, mangrove classification accuracy increased by 7.1% when combining the LiDAR-derived digital terrain model (DTM) and IKONOS (Chadwick, 2011). Species classification accuracy of a temperate forest was raised by 20% by integrating LiDAR elevation metrics and WV-2 image (Zhang and Liu, 2013). Some studies combined airborne hyperspectral images and LiDAR data to discriminate tree species with pixel-based or

object-based approaches and reported to obtain higher accuracy than those produced by using spectral or structural data alone (Jones et al., 2010; Liu et al., 2017).

Multispectral and hyperspectral images only provide two-dimension (2D) data that do not take into account the vertical structure of the forest. Many novel methods were developed to describe the vertical structure and detect the understory with LiDAR data. In some studies, the target classes were firstly defined according to vertical structure characteristics such as understory return density and vertical heterogeneity of the canopy. After that, a series of LiDAR metrics were derived and input to the machine learning (ML) algorithm such as Random Forest for classification. This method was reported to acquire high accuracy between 83% - 88% for understory detection (Martinuzzi et al., 2009; Singh et al., 2015; Wing et al., 2012). Although such classification scheme was able to produce high accuracy, it required prior knowledge to design the target classes and numerous samples for model training, which posed a significant limitation in some study areas where prior knowledge and field data were inadequate. An efficient method to stratify multi-layered forests is to separate LiDAR returns to each layer by analyzing the vertical distribution of point clouds (Duncanson et al., 2014; Hamraz et al., 2017). Basically, LiDAR returns were segmented into objects by the individual canopy or of a certain grid size. The histogram showing the height of LiDAR returns for each object was derived, based on which the troughs/valleys were searched and used to determine the height threshold to stratify different canopy layers. Finally, returns were divided into different canopy layers by identified height thresholds. The procedure did not make a priori assumptions about the shape of the canopy vertical profile. In addition to understory detection, canopy parameters such as crown closure and crown density could be calculated subsequently.

The advance in machine learning classifiers such as Support Vector Machine (SVM) and Random Forest (RF) play an important role to acquire targeted accuracy for vegetation classification (Ballanti et al., 2016; Raczko and Zagajewski, 2017). The deep learning algorithm Convolutional Neural Network (CNN), considered as the state-of-the-art method, has achieved high accuracy in tree species classification by using high-resolution remotely sensed data (Fricker et al., 2019; Hartling et al., 2019). For example, Hartling et al. (2019) compared CNN, RF, and SVM to discriminate eight tree species and CNN significantly outperformed RF and SVM with an overall accuracy of 82.6%, compared to around 52% of the other two algorithms. Some other studies combined LiDAR-derived canopy height model (CHM) and optical images using CNN for classification tasks (Fricker et al., 2019; Hartling et al., 2019; Sun et al., 2019). But the effectiveness of applying various LiDAR metrics in the CNN algorithm remains to be explored.

Few studies compared the high-resolution multispectral and hyperspectral data for vegetation classification, therefore, the feasibility of replacing expensive airborne hyperspectral data with economical spaceborne multispectral data remains unknown. The link between the inherent structure of tree species and how it manifests in LiDAR metrics requires further investigation. Furthermore, the compatibility of combining multispectral or hyperspectral images with LiDAR data in vegetation classification needs to be discovered. However, no extensive research discusses these issues, particularly in mangrove forests. Each mangrove community has its own characteristics on species composition and forest structure (Tam and Wong, 2000). An early study recorded the multi-layered structure and composition of mangroves in Hong Kong based on the field survey (Duke and Khan 1993). However, most of the studies only focus on the overstory species classification and none of them map the understory distribution or explore the integration of understory and overstory in the classification procedure. This study takes a multi-layered mangrove stand in Mai Po, Hong Kong as a testing site to explore the feasibility of mapping the understory and classifying the mangrove species by combining WV-3, airborne hyperspectral, and LiDAR data. An efficient overstory and understory stratification procedure is proposed and three classifiers including RF, SVM, and CNN are

compared for accuracy in terms of species classification. The following objectives are expected to achieve through this study: (1) to provide accurate understory distribution and species distribution maps for mangrove stands in Mai Po Nature Reserve; (2) to understand the impact of the understory in species classification; (3) to identify important predictors for mangrove species discrimination; and (4) to compare multispectral, hyperspectral, LiDAR data and their synergy with various algorithms in mangrove species mapping.

2. Materials

2.1. Study area

The study site is the Mai Po Nature Reserve (22°29'N–23°31'N, 113°59'E–114°03'E) which is located at the northwestern part of New Territories of Hong Kong, China (Fig. 1). The climate regime of Hong Kong is subtropical monsoon, which receives abundant solar heat and rainfall with annual precipitation of about 2200 mm. Mangroves in Mai Po is the largest stand found in Hong Kong with an area of 115 ha and includes diverse flora and fauna (Tam and Wong, 2000). The mangrove ecosystem and its surrounding wetlands were designated as Wetland of International Importance under the Ramsar Convention in 1995. Major mangrove species include 7 native species, namely *Kandelia obovata*, *Avicennia marina*, *Aegiceras corniculatum*, *Bruguiera gymnorrhiza*, *Excoecaria agallocha*, *Acrostichum auerum*, and *Acanthus ilicifolius* (Agriculture, Fisheries, and Conservation Department of the Hong Kong SAR Government (AFCD), 2019) and an exotic tropical species *Sonneratia apetala* that propagated from the plantation mangrove in Futian National Nature Reserve, Shenzhen. The core area in Mai Po Nature Reserve with an area of about 2 km² was selected as the study area (Fig. 1).

2.2. Remotely sensed data and preprocessing

A level 2A WorldView-3 (WV-3) image of the study area was acquired on 20 December 2017 (Table 1), which was atmospherically corrected to normalized surface reflectance values using the AComp method. The WV-3 image was resampled to 0.5 m for the panchromatic image and 2.0 m for the 8-band multispectral image. The WV-3 image was initially projected using the World Geodetic System (WGS) 1984 datum and the Universal Transverse Mercator (UTM) at zone 50 N. Since one of the objectives is to compare the classification performance between WV-3 and hyperspectral images, WV-3 multispectral and panchromatic images were fused by Gram-Schmidt Pan Sharpening method in ENVI 5.3 to obtain the 0.5 m fused image. Both the

Table 1
Specifications of WV-3 image, HSI, and LiDAR data.

Data	Capture time	Specification detail	Spatial resolution
WorldView 3 (WV-3)	Dec 20th, 2017	8 multispectral bands: coastal blue, blue, green, yellow, red, red edge, NIR1, NIR2	2 m (Multispectral) 0.5 m (Panchromatic)
Airborne hyperspectral images (HSI)	Mar 14th, 2019	271 bands, wavelength range: 400–1000 nm, FWHM: 6 nm	0.27–0.38 m (multiple strips)
Airborne LiDAR data	Mar 21st, 2018	X, Y, and Z coordinates, 1–5 return point cloud, 16 bits of return intensity	Return density: 20 point/m ² Footprint diameter: 0.09 m

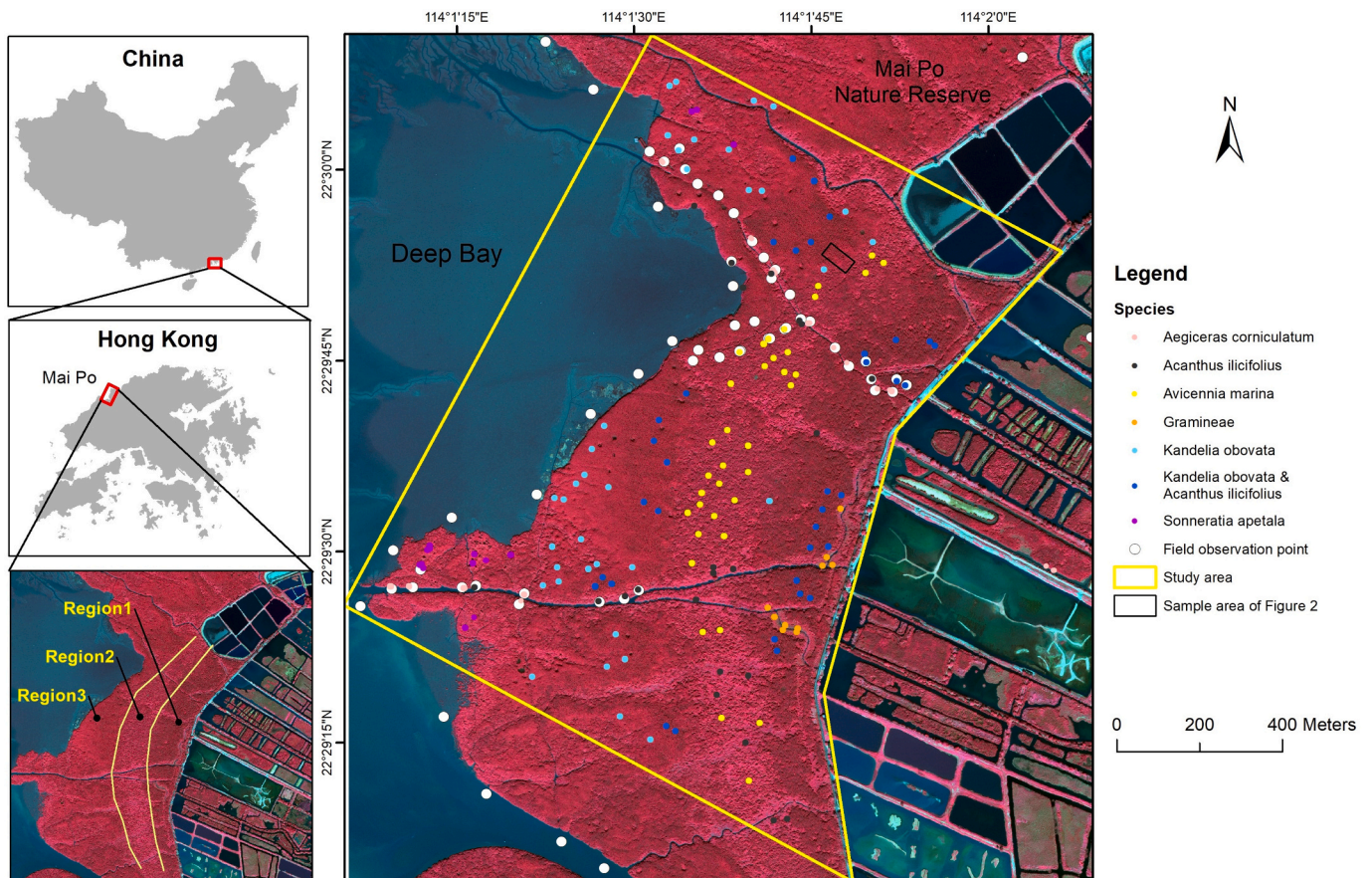


Fig. 1. The study area and the distribution of ground truth locations of seven species. The base map is a pseudo-color WorldView-3 pan-sharpened image composed of NIR1, green, and blue bands. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

multispectral and pan-sharpened (PS) images were reprojected and registered to the Hong Kong 1980 Grid projection in ArcMap 10.6. Areas with Normalized Difference Vegetation Index (NDVI) less than 0.39, which was determined by trial-and-error and visual interpretation were identified as non-vegetated areas and masked out for further analysis.

High-resolution airborne hyperspectral images (HSI) were acquired using the headwall Nano-Hyperspec push-broom sensor. The flight mission was conducted on 13 March 2019 between local time 12:30 and 13:20. The stable weather with sufficient sunlight and low cloud cover minimized the shadow effect and uneven illumination condition. At the time of data acquisition, a low tide level of 1.7–2.3 m at Tsim Bei Tsui, the nearest tide station, was recorded so that background reflectance signals from water could be reduced. Complying with local air traffic control, the flying height was designed as 300 m above ground level in order to obtain high-resolution images. A total of 21 hyperspectral strips with 30% of side overlap were acquired to cover the study area. Raw DN was covered to radiance images which were then calibrated to ortho-images in WGS 1984 geography coordinate in SpectralView software (Headwall Photonics Inc., Massachusetts, USA). Radiometric calibration was processed with the radio calibration file, which recorded the corresponding exposure time and dark current reference. Orthorectification was then conducted using information generated from the inertial motion sensor and DEM. The image area was covered mainly by mangrove trees, where a limited number of tie points ranging from 4 to 15 pairs were manually selected in each strip for registration. The strips were then registered to HK 1980 Grid coordinate system in ArcMap with positional accuracy of 1 m. The initial resolution of the HSI varied from 0.27–0.38 m due to slight difference of flying height which varies in the strips. All registered images were resampled to 0.5 m resolution images with the nearest neighbor method, which retains the original spectral information. Non-vegetated areas were masked out by NDVI less than 0.30. The hyperspectral strips were processed and analyzed individually instead of generating a mosaic due to the large data volume of 111 GB for 21 strips.

Discrete aerial LiDAR data was acquired on March 21, 2018, during a period of low tide level (1.5–1.8 m) using the GL-70A survey-grade LiDAR system. The system was equipped with Riegl VUX-1LR laser scanner (RIEGL Inc., Horn, Austria) which emitted NIR 905 nm with beam divergence at 0.5 mrad for data collection. The flight height was kept between 155 and 270 m to acquire the high-density point cloud. The LiDAR data was projected to HK 1980 Grid, containing attributes of X, Y, and Z coordinates, return intensity, and return number. We conducted a secondary preprocessing to align the LiDAR data with the digital aerial orthophotos (Lands Department, the Hong Kong Government) and to classify the ground point with the filtering method of Progressive Triangular Irregular Network (TIN) Densification (Axelsson, 2000). Following this, the height of returns was normalized by subtracting ground TIN from LiDAR points cloud. The above preprocessing was conducted in LASTools (<https://rapidlasso.com/lastools/>). Fig. 2 shows the cross section of the vertical profile of mangroves in point clouds. Table 1 summarizes some specifications of the three datasets. Based on long-term observation in the past, mangrove species composition was stable within the study areas, but some changes occur in fringe areas (region 3 in Fig. 1) with pioneer species expanding seawards (AFCD, 2019; Chung and Territories, 2013; Duke and Khan, 1993). Besides, by reviewing the records, there were no major incidences that caused significant impacts to the mangrove habitats during the study period. The acquired data in different years are representing the normal conditions of Mai Po.

2.3. Field survey and reference data collection

The species survey was carried out between March and May in 2018 when *Aegiceras corniculatum* (AC) blossomed because flowers of AC improved the certainty and efficiency to differentiate AC from other species. Field accessibility was poor and dangerous due to dense thorn

undergrowth and swampy mudflat. The mangrove stand was accessed along the boardwalk and observed by boat from south to north along the mangrove fringe at high tide level. The field observation location is shown in Fig. 1 and the distance between two observation points is no less than 10 m. Records of each observation points included species, field photos, nearby landmark information, and GPS location within ± 3.65 m positioning error (Garmin, Montana 600).

Our field observation identified three arbor species in the study area, including *Kandelia obovata* (KO), *Avicennia marina* (AM), and the exotic species *Sonneratia apetala* (SA), one shrub or small trees of *Aegiceras corniculatum* (AC), as well as shrub *Acanthus ilicifolius* (AI). KO and AM are the two dominant overstory species with specific area coverage in the study area. KO patches grow along the landside (region 1 in Fig. 1) and seaside areas (region 3 in Fig. 1), and the inner areas between them (region 2 in Fig. 1) are occupied by clusters of AM (Duke and Khan, 1993). Some AC are pioneer species that grow at the edges of the seaside and along the waterway, while some of them scatter among KO and AM. SA is sporadically found along coastal edges. AI, another pioneer species, often distributes along the coastal edges. In particular, this shrub widely spreads as the understory beneath the arbor species. A small area of *Gramineae* plants is found around the riverside. Particularly, two types of KO were identified due to their distinction in appearance and spectral response. One was classified as pure KO having dense crown but with none or sparse understory beneath. Another was noted as the mixture of KO and AI (KOAI), which is characterized by scattered crown coverage of KO with thin foliage and the extensive and lush understory of AI.

Due to limited manpower in the relatively harsh environment for fieldwork, extra samples were collected by visual interpretation from WV-3 PS image aided by fieldwork data. Dominant species KO and AM usually grow in large patches, and this situation was relatively tolerable for the GPS positioning error. While species such as AI and AC in small patches less than 10 m², nearby landmarks were used to help recognize the corresponding patches in the high-resolution image. The understory detection result (Figs. 6 and 8) referred to selected samples for pure KO and KOAI. In total, 245 sample sites were selected (Fig. 1). The sample sites were randomly partitioned in the ratio of 7:3 as training and testing samples respectively to acquire adequate samples for classifier training, especially for CNN (Abdel-Rahman et al., 2014; Fricker et al., 2019; Wong and Fung, 2014). Species classes and the number of sample pixels for each image are described in Table 2. It should be noted that a few SA samples could be found from HSI because of the regular SA removal campaign.¹

The height of the understory was measured in 51 quadrats from 14th to 30th, January 2019. The quadrats were designed along the boardwalk because of poor accessibility in the forest area. Although quadrats were not evenly distributed in the entire study area, they cover various species composition and structure including sparse AI beneath dense KO, dense or moderate AI under mature KO and AM with thin foliage. The quadrat size was 10 × 10 m and each quadrat was divided into four 5 × 5 m sub-grids. Height readings were taken at five locations within the quadrat: one in the quadrat center and four in the centers of four sub-grids. The five measurements were averaged to represent the height of undergrowth layer in each quadrat.

3. Methods

The multi-layered mangrove stratification and species classification workflow is shown in Fig. 3. The procedure consists of three steps: (1) data preprocessing; (2) overstory and understory stratification; and (3) feature selection and species classification. In particular, the stratification result would be used to select samples for classification. Field data

¹ *Sonneratia apetala* (SA) are fast growing and highly reproductive, so they can easily outcompete native mangrove species. To address this threat, the government removes the SA trees regularly (WWF Hong Kong, 2019).

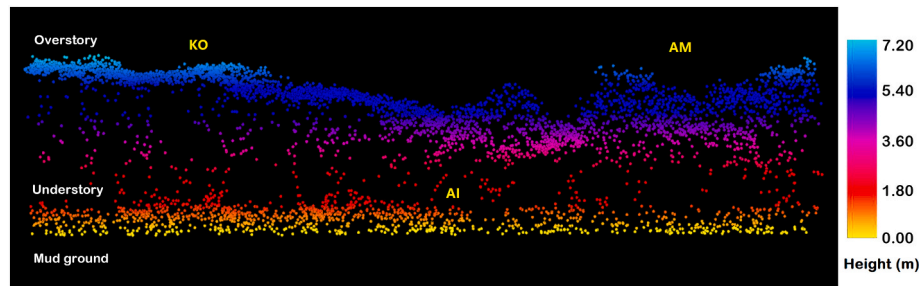


Fig. 2. The illustration of mangrove vertical profile in LiDAR data. The sample area is shown in Fig. 1. The LiDAR point cloud is displayed by height. Different species including *Kandelia obovata* (KO), *Avicennia marina* (AM), and *Acanthus ilicifolius* (AI) and canopy layers are labeled.

Table 2

Vegetation types and the number of sample pixels for training and testing in each image.

Vegetation types	Training pixels			Testing pixels		
	WV-3		HSI	WV-3		HSI
	MS	PS		MS	PS	
<i>Aegiceras corniculatum</i> (AC)	49	219	228	21	89	96
<i>Acanthus ilicifolius</i> (AI)	88	375	373	36	146	145
<i>Avicennia marina</i> (AM)	144	613	613	60	243	243
<i>Gramineae</i> (Grass)	40	171	171	16	65	65
<i>Kandelia obovata</i> (KO)	144	611	611	60	243	243
<i>Kandelia obovata</i> & <i>Acanthus ilicifolius</i> (KOAI)	140	596	596	60	243	243
<i>Sonneratia apetala</i> (SA)	55	235	48	20	90	9
Total	660	2820	2640	275	1119	1044

were applied to evaluate the stratification and classification results.

3.1. Vertical stratification

To stratify the overstory and understory, LiDAR returns within the specific area were delineated to create the height histogram. To this end, segmenting the pixel-based canopy height model (CHM) into objects would be a preferable approach rather than the individual crown delineation in the clustered stand. Applying object segmentation can minimize the internal height heterogeneity within the cluster, which can avoid mixing single-layered canopies and double-layered canopies in the same object because single-layered canopies are mainly found in AI and AC. Since the resolution of CHM could affect the segmentation result, the normalized LiDAR point cloud was used to create the CHM at 0.25 m, 0.5 m, 1 m, and 2 m cell size respectively for multiple segmentation trials. The canopy surface was delineated by Delaunay triangulation from the first returns with a linear interpolation within each triangle. Segmentation was conducted in the eCognition developer 9 (Trimble, CA, USA) with the multiresolution segmentation algorithm. All CHM raster layers were input to segment with the scale parameters ranging from 1 to 20, and the shape and compactness parameters were set to 0.1 and 0.8 respectively. The selection of parameters attempted to determine the best segmentation scale and CHM cell size. The best segmentation result was determined by visual interpretation of segmented objects. Subsequently, for each segmented object, only the above-ground returns were binned into 50 cm intervals to create the height histogram. Local minima of the histogram were searched by detecting where there was a significant decrease to reach a minimum value and then followed by a significant increase across contiguous bins. When multiple local minima were detected, the minimum one not higher than 2.75 m was selected as the threshold to separate understory from overstory. If all local minima were beyond 2.75 m, the minimum was selected as the threshold. As the median of histogram bin of 2.5 m - 3 m, 2.75 m specified here was determined by trial-and-error and fieldwork experiences because understory AI was generally found to be shorter

than 2.5 m. Some histograms revealed areas of single-layered canopy when local minimum was absent. Returns below the defined threshold were classified as understory and returns beyond the threshold were regarded as overstory.

After the return stratification, overstory returns were discarded to generate the CHM of the understory. Understory returns were usually sparser than overstory returns and discontinuous as some areas did not have understory any return. CHM of understory was thus generated in 2 m grid size. Furthermore, in order to understand the relationship between overstory and understory, the proportion of overstory' returns and understory' returns, named as modified canopy cover (Eq. 1) was calculated for each segmented object.

$$\text{Modified canopy cover} = \frac{\text{Overstory returns}}{\text{Overstory returns} + \text{Understory returns}} \times 100\% \quad (1)$$

The vertical stratification and the CHM generation were implemented in the R environment (<https://www.r-project.org/>) with packages "lidR" and "rgdal". Fig. 4 shows the processing flow of vertical stratification.

3.2. Species classification

3.2.1. Feature generation and selection

With the objectives of exploring the effect of spatial resolution for mangrove species identification and comparing the performance of multispectral and hyperspectral images, three optical data WV-3 MS, WV-3 PS, and HSI were used for classification. Given that the spatial resolution of WV-3 MS was coarser than that of HSI, the pan-sharpened WV-3 was able to control the factor of spatial resolution when comparing the performances of multispectral and hyperspectral data. Besides, it was also possible that the pan-sharpening process might reduce the fidelity of original spectra (Li et al., 2017; Li and Qi, 2015), so the radiometric effect of subtle spectra change could also be understood by comparing the classification results of WV-3 MS and WV-3 PS.

The first and last ten wavebands of HSI were removed due to severe noise. As a result, the spectral data consisted of eight broad bands of WV-3 and 251 narrow bands of HSI. A series of LiDAR metrics (Table 3) in five grid sizes ranging from 1 m to 5 m with 1 m interval were calculated respectively based on each sample point to ascertain suitable metrics grid size for mangrove species identification.

To reduce data dimension and unveil relevant data, especially in the HSI data, a wrapper feature selection method, recursive feature elimination (RFE) based on the RF model was applied before the classification. Wrapper methods usually give high accuracy because a large number of models are evaluated during each step of variable add-drop (Fassnacht et al., 2014). RF was able to rank the importance of predictors by Mean Decrease in Accuracy (MDA) that was generated from the out-of-bag error calculation. The RFE-RF entered all features to fit a Random Forest model at the beginning and ranked the predictors. Top-ranked predictors were retained to train the model at each iteration. All

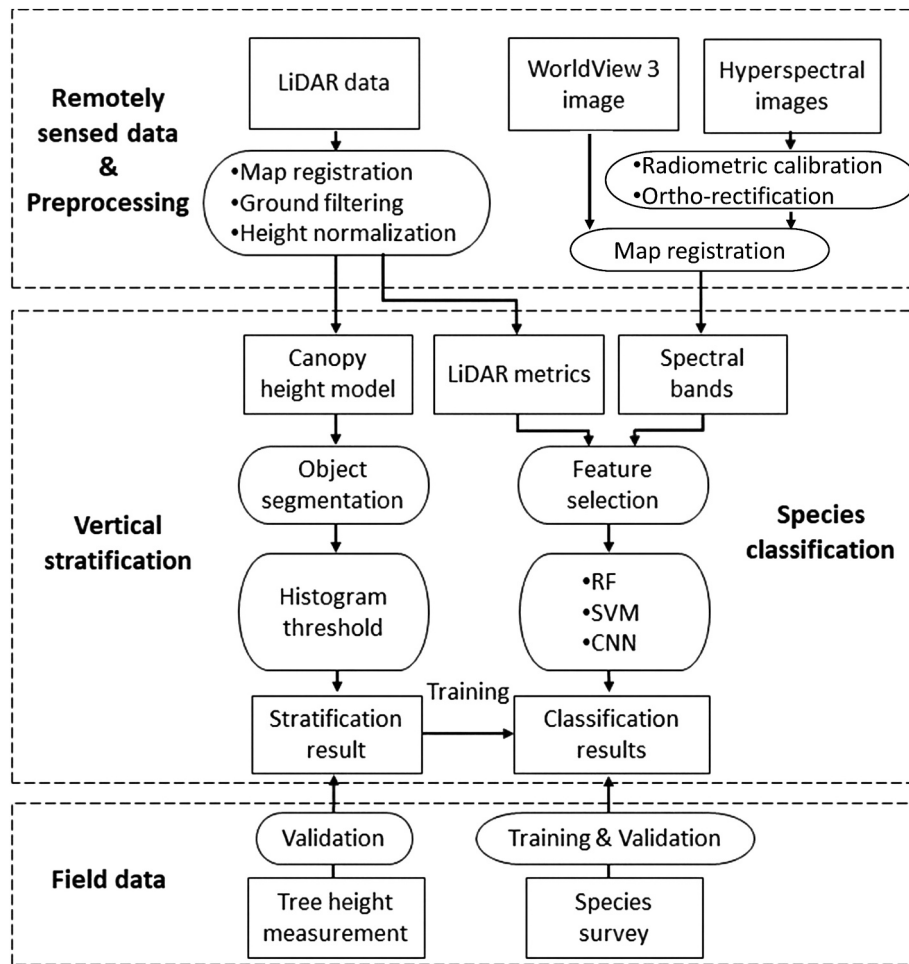


Fig. 3. The workflow proposed for multi-layered mangrove stratification and species classification.

iterations were summarized to a performance profile so that an optimal subset size and important predictors were determined. Previous studies have successfully applied RFE-RF to select significant features and to determine optimal subset size from multi-source data, e.g. optical images and LiDAR data (Ghosh and Joshi, 2014; Li et al., 2019). The 10-fold cross-validation was applied to training samples, and the overall accuracy and Kappa coefficient were used as performance metrics for the RFE-RF process. Feature selection was conducted with R packages of “caret” and “randomForest”.

3.2.2. Classification techniques

The pixel-based classification was conducted using three algorithms CNN, RF, and SVM for overstory species identification. A pixel-level CNN that considered the contextual spatial information can obtain high classification accuracy with WV-2 or hyperspectral image (Fricker et al., 2019; Hartling et al., 2019). A CNN structure was adapted from Fricker et al. (2019) and the model structure is shown in Fig. 5. An input patch was defined as $n \times n \times d$ data cube, where n was the patch size and also the hyperparameter of the CNN model. The depth of cube d referred to the number of feature inputs. Each patch was constructed with a central pixel and its surrounding pixels within the $n \times n$ window. The patch size affected how much of the surrounding context would be used in the network to predict the central pixel which was the patch label. Hereafter, patches went through a cascade of convolutional layers with the kernel (filter) size of 3 and stride of 1 following a 1×1 convolutional layer as output. The input channel of the convolution layer was c . The c of the first layer equaled to the number of image band d , and c doubled at each subsequent layer until the maximum number of channel c' was

reached, where c and c' were hyperparameters. The output channel of the final layer was set as seven which corresponds to the seven species classes. As a result, the number of $(n-1)/2$ of convolutional layers was constructed using Rectified Linear Unit (ReLU) activations as the non-linear activation function. The cross-entropy loss function was used at last to output the probability of each class. Training and testing points were constructed into $n \times n \times d$ patches to assemble training and testing data. The fine-tuning of CNN was based on the parameter setting in the original model. As a consequence, patch size n was set to 15 for the 0.5 m HSI and the WV-3 PS image and 9 for the 2 m WV-3 MS image. A small patch does not contain sufficient spatial context for central pixel prediction. Conversely, a large patch size might include a large number of pixels belonging to other species which may be irrelevant to the central pixel and is also computationally inefficient. The maximum number of the output channel c' was set to 128. Although more filter kernels might improve the classified power, it may cause overfitting to the training data. As a result, 7 convolution layers were constructed for the HSI and WV-3 PS image, while 4 convolution layers were constructed for the WV-3 MS image. A batch size of 100 was used for each training iteration. The stochastic optimization (Kingma and Ba, 2015) was applied to the model by Adam optimizer with learning rate and momentum set as 0.0001 and 0.9 respectively. The model was trained up to 100 epochs and then the model was retrained with the epoch that produced the best classification accuracy. The CNN classification was implemented with PyTorch 1.1.0 (<https://pytorch.org/>) in Python 3.7.3 environment. To our knowledge, high-resolution HSI and LiDAR metrics layers were first explored with the pixel-based CNN classifier for mangrove species classification.

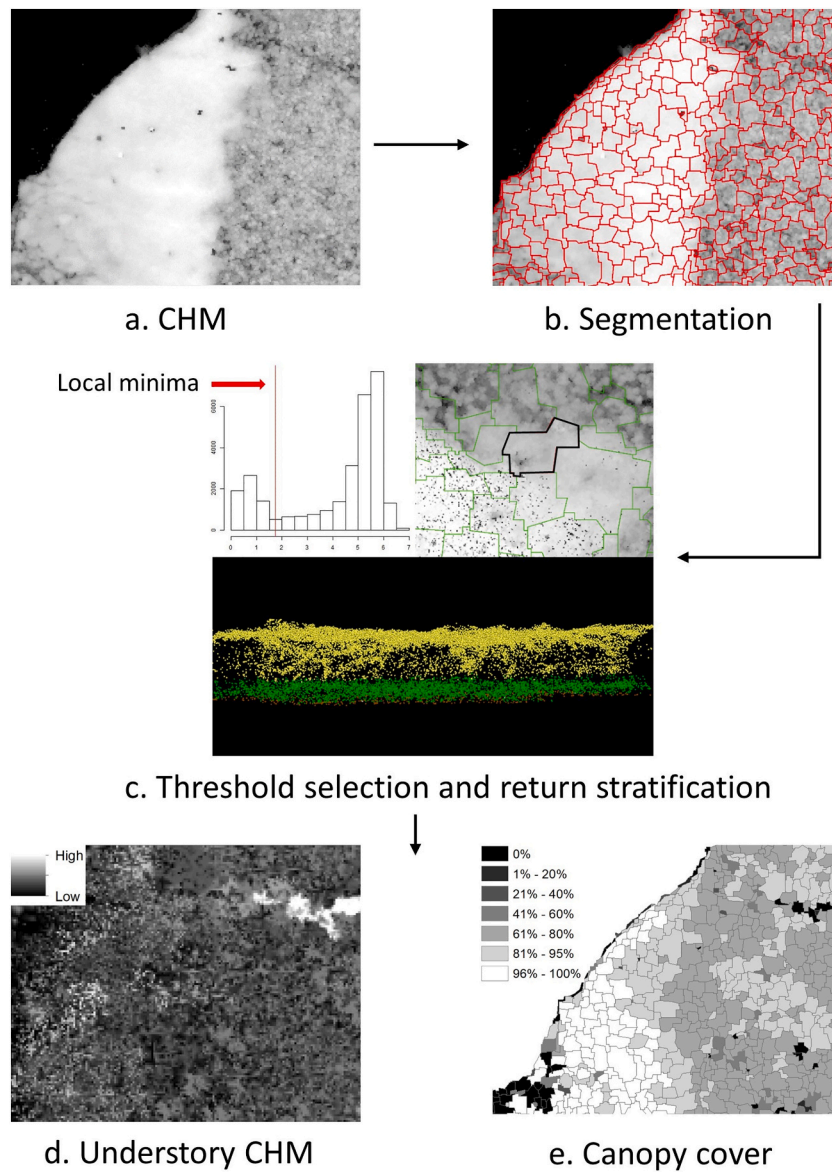


Fig. 4. The processing flow of vertical stratification.

Table 3

A list of LiDAR metrics.

Abbreviation of LiDAR metrics	Metrics details
Return height distribution	
CHM	Canopy height model
zmean	Mean height
zsd	Standard deviation of the height distribution
zskew	Skewness of the height distribution
zkurt	Kurtosis of the height distribution
pzabovezmean	Percentage of returns above the mean height
Percentile of the height distribution	
zqx	Xth percentile (quantile) of the height distribution
(x = 5,10,15...95)	
Vertical structure	
CRR	Canopy relief ratio = $((zmean - zmin) / (zmax - min))$
pzabove2	Percentage of returns above 2 m
cover_o	Overstory canopy cover, the same as modified canopy cover

bootstrap samples originating from the training data were used to grow a large number of decision trees and a subset of input features (Mtry) was randomly selected to search the best split for each node. The final prediction was determined by voting the results of decision trees (Breiman, 2001). 500 decision trees were generated and the parameter Mtry was tuned from 2 to k , where k was the number of input features.

A Radial Basis Function (RBF) kernel based SVM was applied, which better handled reality circumstances (Fan et al., 2005). Support vectors were searched to build the hyperplane with the maximum margin to separate the classes. The performance of SVM was affected by two important parameters: constraint violation (C) that controlled the overfitting and sigma (σ) that shaped the hyperplane. These two parameters were tuned by grid searching within the range of 10^{-3} - 10^{+3} . A 10-fold cross-validation was applied in the RF and SVM model training to determine optimal parameters. RF and SVM were implemented in R with the “randomForest” package and “e1071” package respectively.

3.3. Accuracy assessment

The mangrove vertical stratification result was validated in two

RF assembles many decision trees for classification. Different

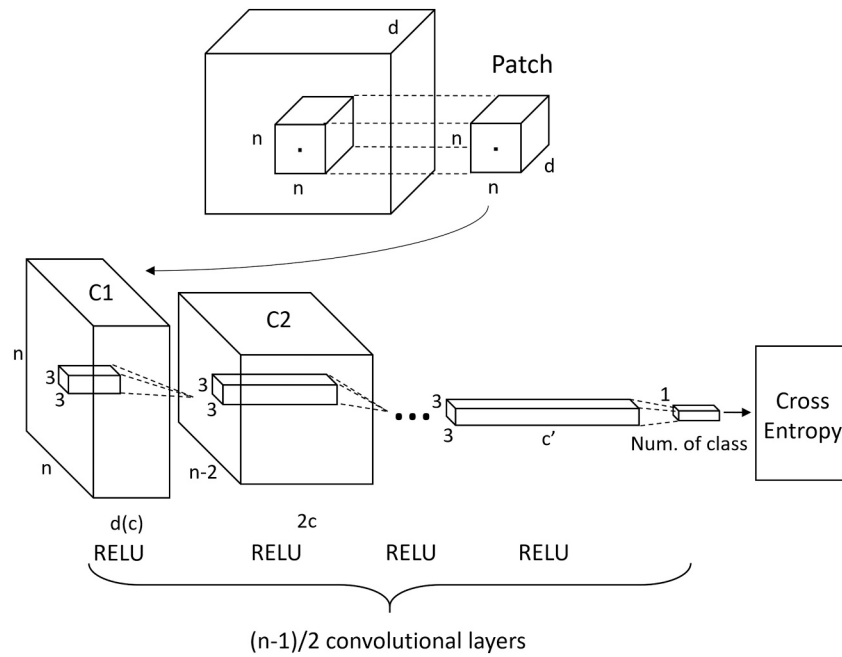


Fig. 5. Pixel-based CNN architecture.

aspects. First, the understory distribution acquired from LiDAR data was checked with the ground truth. Since no understory survey had been conducted in Mai Po before, no available archive about the understory distribution can be used for validation. Therefore, the vertical stratification result was validated by comparing our understory distribution map with the fieldwork experience of many parties e.g. WWF, AFC, and mangrove researchers. Second, the vertical stratification result was assessed by the accuracy of understory canopy height estimation. The estimated understory height would be compared with in situ measured height from the 51 quadrats. The mean value within a window of 5×5 pixels corresponding to the location and size of each quadrat was extracted from the understory *CHM* layer. A Pearson correlation was conducted between the mean height derived from *CHM* and the measured height.

For species classification, independent testing samples were used to quantify the mapping accuracy of species. Confusion matrices were constructed by comparing the predicted class and labeled classes of testing samples. Classification accuracy was assessed by the overall accuracy (OA) and Kappa coefficient, whilst the accuracy of the individual class was evaluated via the user's accuracy (UA) and the producer's accuracy (PA). The disagreement between the predicted map and the reference map was also evaluated. Two measures known as the quantity disagreement (QD) and allocation disagreement (AD) that decompose the overall level of disagreement were calculated. The QD is the absolute dissimilarity between the number of reference samples and the predicted ones, while the AD reflects the differences between the reference and the predicted maps caused by the less than the maximum match in the spatial allocation of the classes (Pontius and Millones, 2011).

4. Results

4.1. Vertical stratification

After many attempts of segmentation, the best segmentation result derived from the *CHM* layer with the scale parameter of 5 was selected, which divided the study area into 3095 objects as shown in Fig. 6. Most objects had a threshold of 1.75 m (0.314 km^2) and 2.25 m (0.277 km^2). 587 objects (0.186 km^2) were identified with no undergrowth layer, and a small amount of area had a high threshold higher than 2.75 m (0.053

km^2). Objects were categorized by modified canopy cover coupled with information in field observation and visual interpretation of LiDAR returns of individual objects (Fig. 6). The study area mainly includes four distinctive types of vertical strata with examples for each type shown in Fig. 7. The statistical summary of different vertical strata is described in Table 4.

Areas with 0% of overstory return (in white color) had no threshold found and were identified as the single-layered low plants. Areas had less than 20% overstory returns (in beige color) because those objects were open canopy areas including some returns from surrounded tall trees such as KO and AM. Areas with 0–20% of overstory returns are essentially considered as the first vertical strata type, the single-layered shrub (Fig. 7 a). These areas were usually distributed along river channels or waterways dominated by pioneer species of AI and a small amount of *Gramineae* plants. Aside from the above circumstance, other areas were the convex part of the seaward mangrove fringe (the south part in region 1) originating from the mangrove seaward expansion since 1969 where young and dense AC and KO dominated (Duke and Khan, 1993). Areas with 20% - 60% overstory returns (in yellow and orange color) were regarded as the second vertical strata type (Fig. 7 b), namely open or less dense overstory but with dense undergrowth beneath. Parts of these areas were occasionally found near the seaward fringe or riverside where individual mature trees such as SA, KO, and AC occurred above the dense AI layer. While other parts of these areas locating near the landside or in the middle of the stand (region 1 and 2) were usually low-density mature KO or AM, which was consistent with early survey results (Duke and Khan, 1993). Areas with 60% - 95% overstory returns (in light green and green color) accounted for the majority part of the mangrove stand which was treated as the third vertical strata type (Fig. 7 c). Mature AM and KO with moderate crowns dominated the upper canopy with lush thickets of AI beneath. Areas with denser overstory canopy (80% - 95% overstory returns, in green color) were often found toward the sea fringe when compared to those with moderate overstory canopy (60% - 80% overstory returns, in light green color). Areas having over 95% of overstory returns (in blue color) were classified as another single-layered canopy vertical strata type (Fig. 7 d). These areas originated from the seaward mangrove expansion, which were usually composed of dense and relatively young KO but with minimal or even absence of understory. Since mangrove stand

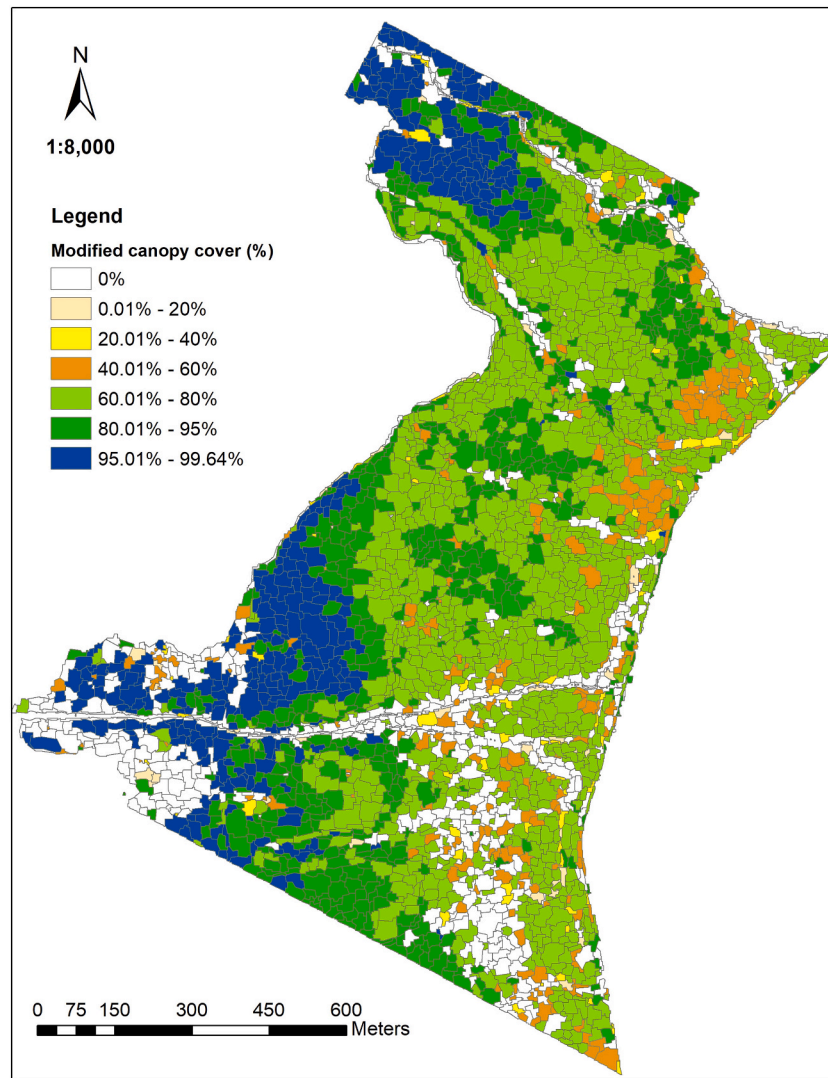


Fig. 6. Map of the modified canopy cover for segmented objects.

continuously expanded seaward toward the Deep Bay (Chan et al., 2011; Duke and Khan, 1993), the age of mangrove trees decreased from the inner area to the seaward fringe. Young mangroves often had dense crowns because self-thinning process did not occur (Khan et al., 2013; Khan and Kabir, 2017). Dense crowns blocked the sunlight to reach the ground that could limit the presence and growth of the understory. Apart from sunlight, the mudflats of low-lying areas around the seaside were softer than those in the inland areas, which tend to restrict the shrub species to root steadily.

Fig. 8 shows the CHM of overstory and understory in 2 m grid size. Generally, the overstory canopy height showed a bimodal form when moving from the landside (region 1) to the seaside (region 3) because the middle zone (region 2) was dominated by mature AM which were often shorter in height comparing with the mature KO distributing on both sides. The understory canopy height decreased toward the seaside. Understories between 1 m and 1.5 m had the largest areal coverage of 0.352 km² which were mainly distributed in inland and middle areas (region 1 and region 2) under the mature overstories. The thinning process of mature KO and AM matched with the observation of large gaps in the canopy, which allowed understories to thrive under plenty of sunlight. While short understories below 0.5 m were frequently found in the northern and southern seaward fringe (region 3) beneath those young and dense overstories. Specifically, a small number of undergrowth with about 2–3 m height was found beneath tall and mature KO

with about 8–10 m height near the inland areas (region 1).

Fig. 9 shows that measured and estimated understory heights were significantly correlated, with a correlation coefficient of 0.71 (p -value < 0.001). It was noted that most of the understory height beneath the AM was underestimated. While a few large outliers occurred in the estimated understory height under the KO.

4.2. Species classification

Table 5 lists the optimal feature sets of seven input data determined by RFE. Visible bands of WV-3 MS and WV-3 PS were the top-ranking features. Fig. S1 compares the spectral differences of training samples obtained from WV-3 MS across the seven species using boxplots. The blue, green, yellow, and red bands which are sensitive to the concentration of pigments such as chlorophyll, carotenoids, and xanthophyll in leaves, were able to separate grass, AM, and SA (Blackburn, 1998; Féret et al., 2017; Stagakis et al., 2014). Even though AC and AI shared similar spectral values, the green and yellow bands could separate them to some degree. KO often had higher values than KOAI in the red edge and the two NIR bands, which is related to the multiple scattering within the leaf structure.

Boxplots in Fig. S2 compare the differences among the seven species in twenty selected hyperspectral wavelengths. Selected wavelengths from HSI mainly consisted of the red edge region between 682 nm and

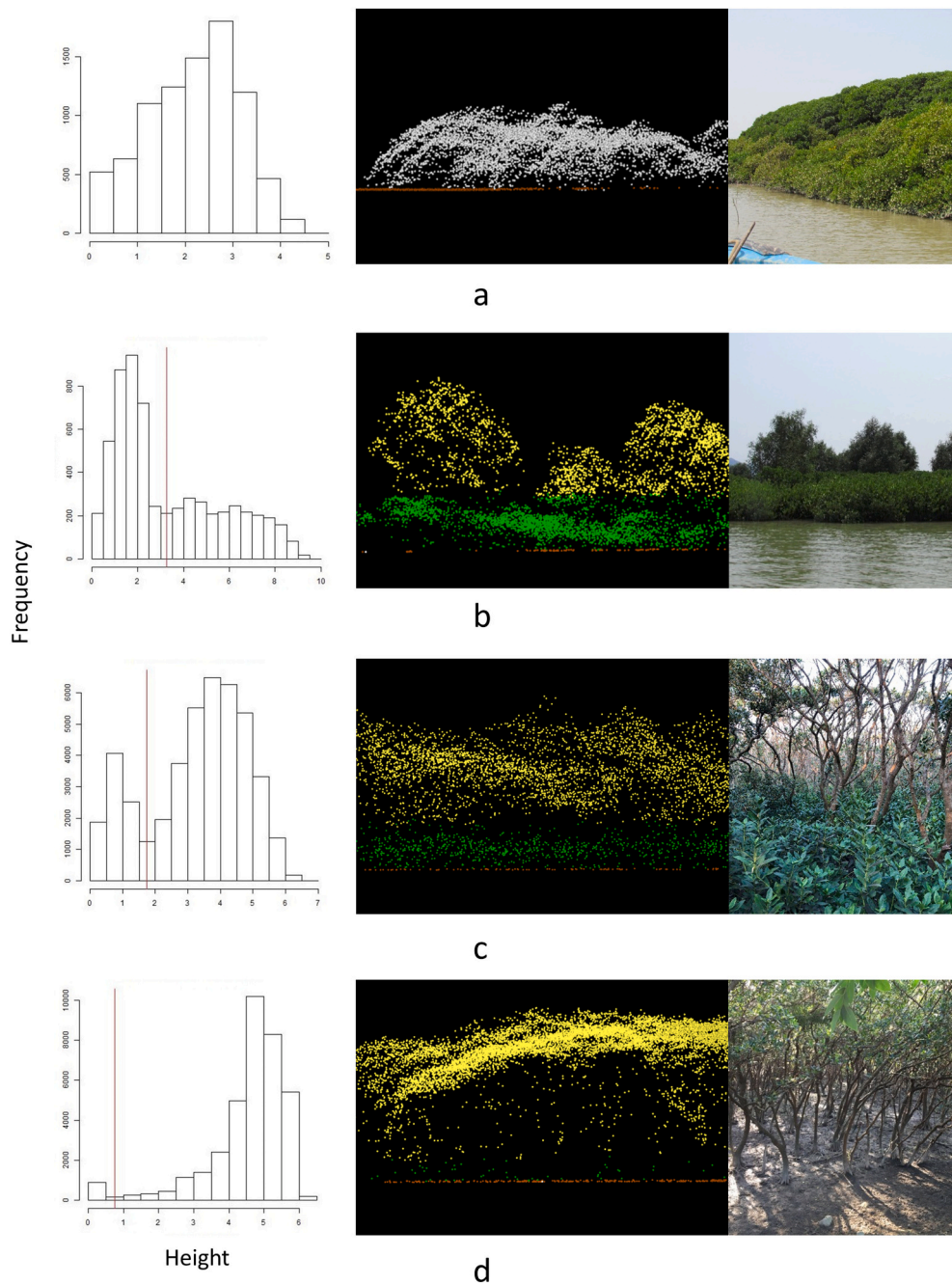


Fig. 7. Examples of object's height histogram and the selected height threshold (red line), the vertical profile of LiDAR returns (white point: single-layered vegetation, yellow point: high vegetation, green point: low vegetation, brown point: ground), and field photos. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

A summary of the modified canopy cover for objects.

Modified canopy cover (%)	Number of objects	Total area (km ²)	Description
0%	587	0.186	Single-layered shrub
0.01%–20%	50	0.011	
20.1%–40%	79	0.016	Less dense overstory, dense understory
40.1%–60%	255	0.077	
60.1%–80%	1116	0.501	Moderate overstory, moderate understory
80.1%–95%	654	0.285	
95.1%–99.6%	354	0.173	Dense overstory, sparse and low understory or no understory
Total	3095	1.248	

744 nm. Other selected wavelengths were located at the NIR region between 769 nm and 929 nm and at the green region at 522 nm. Wavelengths of 682 nm and 689 nm could significantly differentiate KOAI and SA from other species. Longer wavelengths from 695 nm to 706 nm could identify grass, for example, some *Gramineae* in yellow-green color. The spectral value of KOAI was much higher than KO when wavelengths were shorter than 702 nm. Wavelengths from 720 nm to 784 nm were able to identify AC, AI, and grass. NIR wavebands from 873 nm to 929 nm were able to differentiate AC from KO to some degree.

Fig. 10 describes the RF classification accuracy varying with LiDAR metrics in different grid sizes. The overall accuracy significantly increased from the grid size of 1 m (0.736) to 2 m (0.780) and slightly increased from 2 m to 3 m (0.804), but no increase was observed with

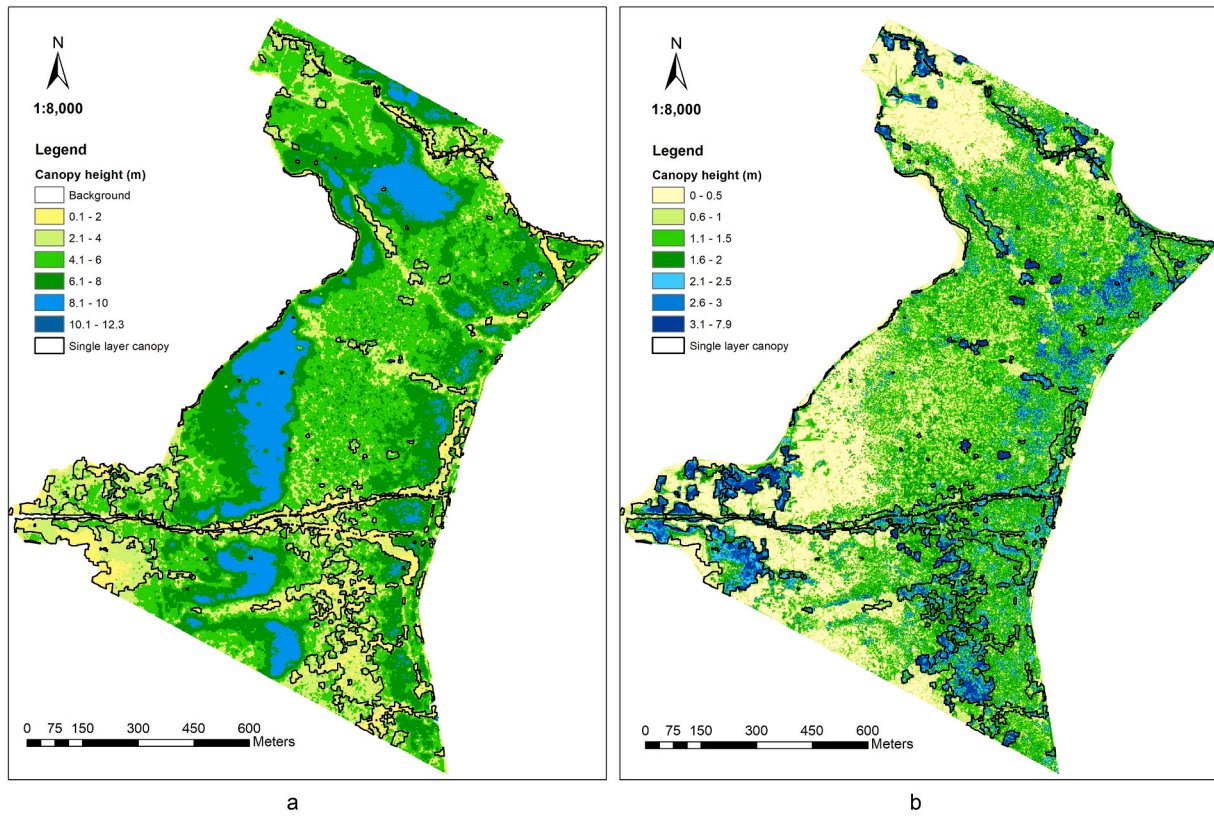


Fig. 8. a: map of canopy height of the overstory; b: map of canopy height of the understory.

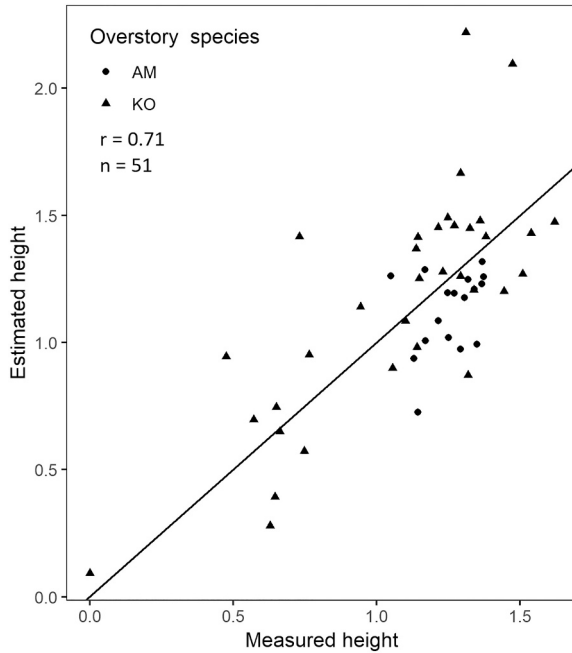


Fig. 9. Comparison of measured understory heights and estimated understory heights.

the grid size larger than 3 m. In terms of the PA for individual species, the 2 m LiDAR metrics generated similar accuracy in identifying AM, KOAI, KO, and grass compared to the 3 m metrics, but it performed even better in locating AI because finer resolution metrics could offer detailed description for species with small canopy such as AI and grass. Therefore, the 2 m grid size was eventually chosen to compute the LiDAR

Table 5

Feature selection results of different data source.

Data	Resolution (m)	Optimal feature size	Selected feature (Listing in descending order of importance)
<i>Single-source data</i>			
WV-3 MS	2	7	B3, B4, B6, B5, B7, B8, B2
WV-3 PS	0.5	7	B6, B3, B5, B4, B7, B8, B2
HSI	0.5	20	p704, p744, p695, p700, p698, p720, p702, p784, p726, p929, p689, p769, p522, p682, p905, p706, p724, p898, p873, p735
LiDAR	2	10	CHM, zsd, CRR, zq95, zq5, zq10, pzabovezmean, zq20, zq90, zskew
<i>Combination data</i>			
WV-3 MS + LiDAR	2	13	CHM, zq95, zq90, zsd, B3, B4, B6, zq20, B5, B2, pzabovezmean, B7, B8
WV-3 PS + LiDAR	0.5	17	zsd, CHM, zq95, zq90, zq20, CRR, B6, B7, pzabovezmean, zskew, B8, B3, zq10, zq5, B4, B5, B2
HSI + LiDAR	0.5	21	CHM, zsd, zq95, zq20, CRR, zq90, p682, zq10, zq5, pzabovezmean, p726, zskew, p724, p735, p769, p689, p744, p522, p700, p695, p698

p704 = wavelength in 704 nm, etc.

metrics with the consideration of both classification accuracy and ability to capture small canopies.

Fig. S3 shows differences among the seven species in 10 selected LiDAR metrics. Selected LiDAR metrics describing the overstory canopy height including CHM, zq95, and zq90 were useful to separate most of the species because of their intrinsic height difference. For instance, KOAI are often taller than KO, followed by AM and AC. As for species with similar canopy height, they could be further differentiated by LiDAR metrics that described the vertical structure. LiDAR metrics

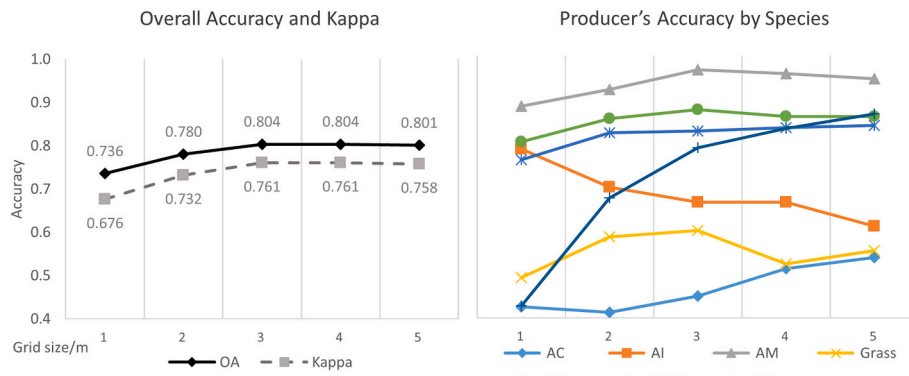


Fig. 10. Random Forest classification accuracy of different LiDAR metrics grid sizes.

reflected the crown shape like *CRR* and canopy cover like *pzabovezmean* were useful to identify single-layered KO. Metrics *zskew* and *zsd* describe return distribution of overstory and understory which were good indicators to differentiate multi-layered AM and KOAI.

Overall classification results produced by RF, SVM, and CNN classifiers with standalone WV-3, HSI, and LiDAR features or the combination of them were compared in Table 6. When using RF and SVM classifiers, WV-3 MS generated higher accuracy (0.70–0.72) than WV-3 PS and HSI (0.63–0.65). LiDAR data produced better overall accuracy (0.78–0.79) than optical images (0.63–0.72) with RF and SVM. CNN significantly improved the overall accuracy of optical images to 0.78–0.82 but generated similar OA with RF when using LiDAR data. The synergy of spectral and LiDAR features achieved the highest overall accuracy, three classifiers generated similar OA reaching 0.86–0.88 and 0.84–0.89 when combining LiDAR metrics with HSI and WV-3 respectively.

In terms of algorithms, differences in QD and AD of the RF and SVM were not significant. RF and SVM produced similar ranges of QD (0.06–0.10) and AD (0.14–0.29) when using single-source data. By combining optical and LiDAR, RF and SVM reduced the QD to 0.03–0.08 and AD to 0.06–0.13. CNN produced similar QD compared to the other two algorithms when processing WV-3 MS data, while it produced lower QD (0.03–0.06) with PS data. Notably, CNN generated the lowest AD (0.06–0.14) among the three algorithms when the same data source was used, which reflected CNN was robust in reducing the spatial disagreements. The confusion matrices of each classification result can be found in the supplementary data.

Fig. 11 shows the resultant species distribution map produced by combining WV-3 MS and LiDAR data using the RF classifier (OA = 0.86, Kappa = 0.83, QD = 0.08, QA = 0.06). The result was generally consistent with the distribution pattern observed in the field. Most of AM were distributed in the central area. Mature KO with dense AI beneath (defined as KOAI) were found on both sides of AM. Young KO with dense crown spread toward the sea. AI and AC were mostly situated along the waterways and sea edges. Besides, a large areas of pioneer AC was

emerging in the southwest, and some isolated AI patches were also found in the southern part of the study area.

Since RF and SVM delivered similar results using different data sources, RF was selected to compare the PA and UA of the seven species with CNN as shown in Table 7. Fig. 12 compared RF and CNN classified maps in an area of high species diversity. The location of this area is indicated in Fig. 11.

Large variation in accuracy was observed across the seven species and also when different data sources and classifiers were concerned. AC often obtained the lowest accuracy among all species as they were frequently misclassified as KO, AM, and AI. When WV-3 MS was input as the single data and classified with RF, AC yields PA of 0.19 and UA of 0.4. The results were even poorer with PA of 0.1 and UA of 0.23–0.36 when using WV-3 PS and HSI. LiDAR data reduced the confusion of short AC and relatively tall KO and both PA and UA of AC were improved to 0.43 and 0.53 respectively. The PA of AC was improved to 0.24–0.51 using CNN algorithm and to 0.37–0.68 when using combined data. However, CNN appeared to overestimate AC along the waterside as shown in Fig. 12 i, j, k, and l. The distribution of shrub species AI and grass were mapped accurately with PA falls between 0.61 and 0.94 for AI and between 0.50 and 0.98 for grass. While the distribution of KO, KOAI, and AM was better described by LiDAR data and HSI when compared to the two WV-3 images even though the latter produced relatively higher accuracy. As shown in Fig. 12 a, b, and c, WV-3, HSI, and LiDAR data distinguished AI and grass from other species clearly. However, areas of KO and KOAI were indiscernible using WV-3 data. KOAI, KO, and AM could be identified effectively in HSI because KOAI had apparent canopy gaps and AM looked darker than KO due to their thinner foliage over thick branches. These three species also could be differentiated by canopy height derived from LiDAR data. Since the same species often congregated together to form large patches, CNN improved the accuracy of KO to 0.83–0.98; of KOAI to 0.80–0.94; and of AM to 0.77–0.96 by reducing the scattered pixels and with clear boundaries. RF produced high accuracy for SA by using WV-3 MS

Table 6

Overall classification results of different data sources produced by RF, SVM, and CNN classifiers.

Data	Classifier											
	RF				SVM				CNN			
	OA	Kappa	QD	AD	OA	Kappa	QD	AD	OA	Kappa	QD	AD
WV-3 MS	0.70	0.63	0.08	0.22	0.72	0.66	0.08	0.20	0.78	0.75	0.09	0.13
WV-3 PS	0.64	0.56	0.10	0.29	0.65	0.57	0.07	0.28	0.82	0.80	0.04	0.14
HSI	0.63	0.54	0.09	0.28	0.65	0.55	0.06	0.29	0.82	0.80	0.04	0.14
LiDAR	0.78	0.73	0.06	0.15	0.79	0.74	0.08	0.14	0.80	0.76	0.08	0.12
WV-3 MS + LiDAR	0.86	0.83	0.08	0.06	0.84	0.81	0.05	0.11	0.88	0.85	0.06	0.06
WV-3 PS + LiDAR	0.84	0.80	0.03	0.13	0.84	0.80	0.03	0.13	0.89	0.86	0.03	0.08
HSI + LiDAR	0.87	0.84	0.04	0.09	0.86	0.83	0.03	0.11	0.88	0.85	0.06	0.06

OA = Overall accuracy, QD = Quantity disagreement, AD = Allocation disagreement.

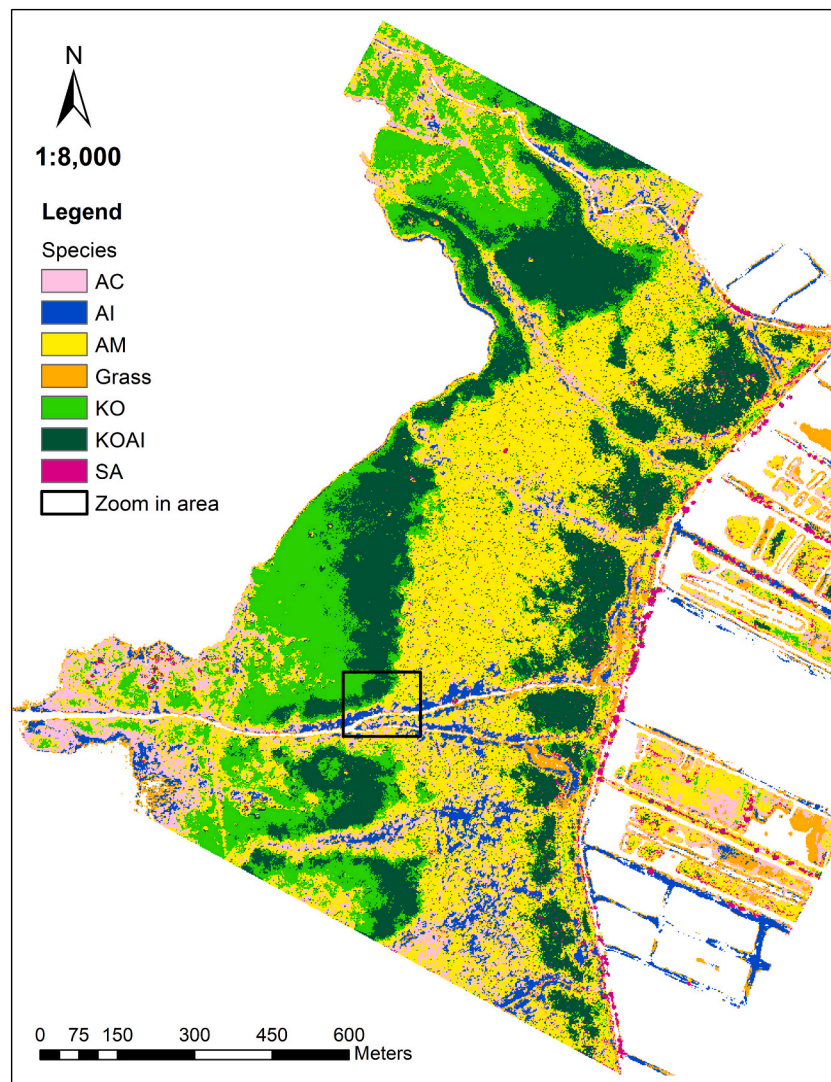


Fig. 11. Random Forest classified map of mangrove species distribution produced by WV-3 MS and LiDAR data.

Table 7

A comparison of the classification results produced by RF and CNN for all species.

Data	Classifier	Species PA/UA							OA
		AC	AI	AM	Grass	KO	KOAI	SA	
WV-3 MS	RF	0.19/0.40	0.64/0.77	0.63/0.62	0.81/0.93	0.83/0.74	0.78/0.64	0.82/0.95	0.70
	CNN	0.24/1.00	0.67/0.79	0.62/0.61	0.94/1.00	0.85/0.85	0.80/0.81	0.82/0.90	0.78
WV-3 PS	RF	0.10/0.23	0.61/0.68	0.53/0.54	0.58/0.86	0.76/0.69	0.79/0.59	0.40/0.54	0.64
	CNN	0.43/0.58	0.75/0.83	0.77/0.75	0.91/0.97	0.96/0.84	0.86/0.84	0.83/0.91	0.82
HSI	RF	0.10/0.36	0.72/0.64	0.62/0.58	0.80/0.78	0.66/0.56	0.65/0.95	0.33/1.00	0.63
	CNN	0.51/0.55	0.96/0.87	0.77/0.80	0.77/1.00	0.94/0.84	0.84/0.85	0.00/0.00	0.82
LiDAR	RF	0.47/0.53	0.86/0.82	0.93/0.76	0.50/0.62	0.77/0.85	0.83/0.89	0.68/0.71	0.78
	CNN	0.47/0.56	0.72/0.76	0.92/0.80	0.38/0.46	0.83/0.91	0.87/1.00	0.91/0.59	0.80
WV-3 MS + LiDAR	RF	0.68/0.81	0.83/1.00	1.00/0.76	0.94/0.83	0.78/0.90	0.87/0.87	0.82/1.00	0.86
	CNN	0.37/1.00	0.94/0.85	0.83/0.79	0.69/1.00	0.98/0.95	0.93/0.86	1.00/0.88	0.88
WV-3 PS + LiDAR	RF	0.50/0.57	0.86/0.90	0.94/0.77	0.83/0.90	0.83/0.90	0.88/0.91	0.74/0.82	0.84
	CNN	0.49/0.58	0.84/0.88	0.93/0.90	0.97/0.89	0.96/0.92	0.94/0.99	0.87/0.81	0.89
HSI + LiDAR	RF	0.53/0.74	0.86/0.84	0.91/0.91	0.98/0.91	0.95/0.85	0.72/1.00	0.89/0.80	0.87
	CNN	0.58/0.89	0.94/0.93	0.96/0.83	0.75/0.94	0.91/0.84	0.91/0.94	0.00/0.00	0.88

PA = Producer's accuracy, UA = User's accuracy, OA = Overall accuracy.

individually with PA of 0.82 and UA of 0.95, but it still confused SA with AI along the waterside. HSI obtained the poorest accuracy for SA with PA of 0.3, which was due to insufficient training samples. Although RF applied with LiDAR data obtained moderate accuracy for SA with PA of

0.68 and UA of 0.71, the overestimation of SA was largely reduced as shown in Fig. 12 h. Comparing with RF, CNN magnified the overestimation of SA along the waterside except using HSI (Fig. 12 o).

We found that WV-3 MS and PS had some differences in classification

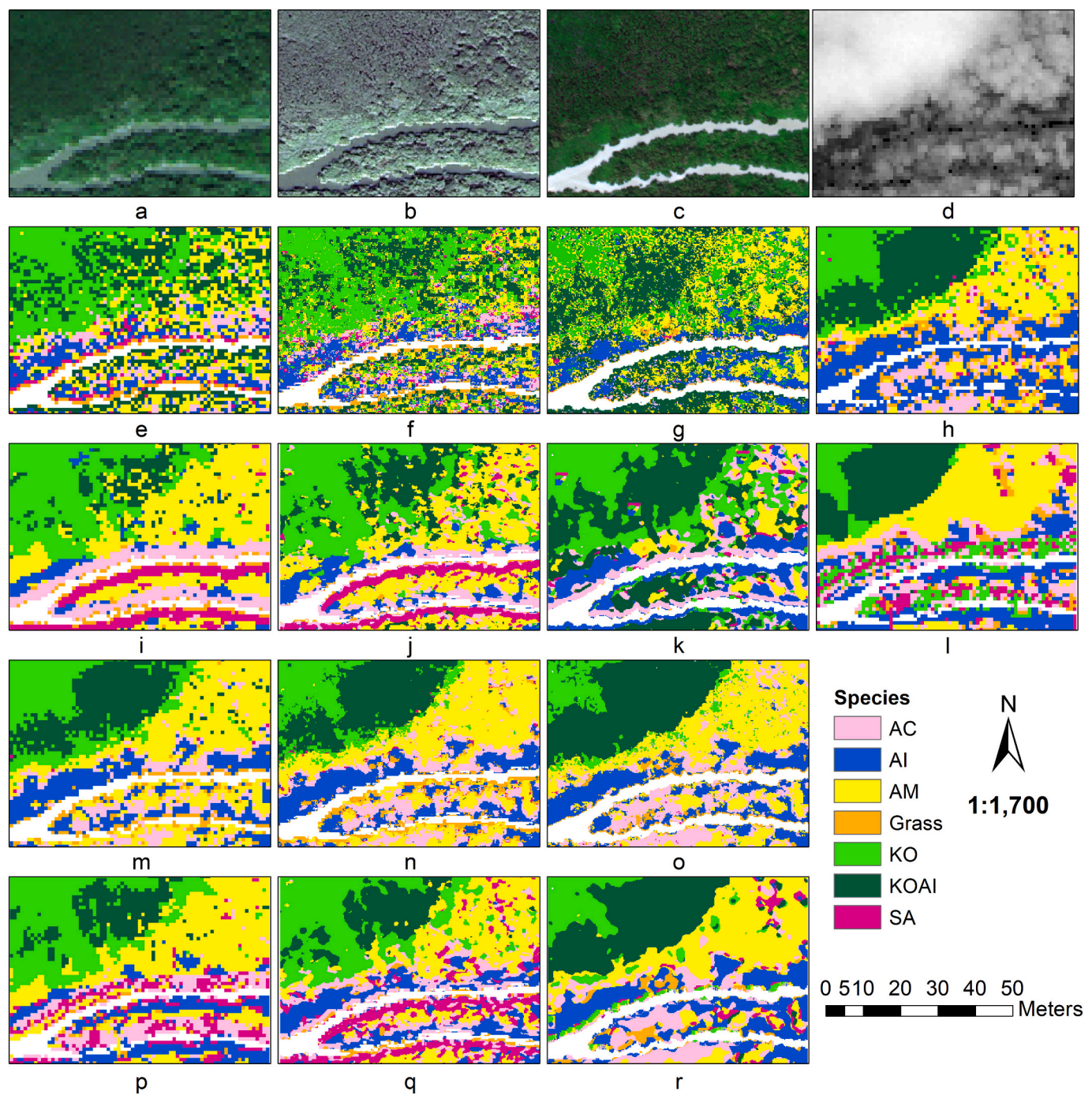


Fig. 12. A comparison of species distribution maps derived from different data combination using RF and CNN classifiers in zoom-in area (of Fig. 11), a: WV-3 MS image in true color, b WV-3 PS image in true color, c HSI in true color, d: 2 m CHM, e: WV-3 MS with RF, f: WV-3 PS with RF, g: HSI with RF, h: LiDAR metrics with RF, i: WV-3 MS with CNN, j: WV-3 PS with CNN, k: HSI with CNN, l: LiDAR metrics with CNN, m: WV-3 MS + LiDAR with RF, n: WV-3 PS + LiDAR with RF, o: HSI + LiDAR with RF, p: WV-3 MS + LiDAR with CNN, q: WV-3 PS + LiDAR with CNN, r: HSI + LiDAR with CNN.

accuracy, especially for those species whose spectral properties were often affected by foliage gaps such as KO, AM, and grass. But WV-3 and PS delivered very similar mapping patterns for species distribution as shown in Fig. 12. WV-3 PS and HSI produced similar overall accuracy and classified disagreement (OD and AD) when they were used alone and integrated with LiDAR data. HSI yielded better accuracy than WV-3 PS for most species, except SA. While these two data produced close classification accuracy and mapping results for dominant species AM, KO, and KOAI.

Combined data obtained higher accuracy for most species than that

of single-source data, and the improvement was especially significant when using RF. RF improved PA of AC to 0.68 when combining LiDAR with WV-3 MS and to 0.50–0.53 when WV-PS and HSI were used. Shrub species AI and grass were well identified by all three data combinations with PA ranging from 0.83 to 0.98. As shown in Fig. 12 m, n, and o, the distributions of KO, KOAI, and AM were clearly depicted and the discrimination between multi-layered AM versus KOAI, and KO versus KOAI was also improved with their PA ranging from 0.71 to 1.00 and UA ranging from 0.76 to 1.00. The confusion between AI and SA was largely reduced with PA of 0.83–0.86 for AI and 0.74–0.89 for SA, and the

overestimation of SA was reduced by combined datasets. When applied CNN with multi-source data, better accuracy was obtained for AI, AC, and KOAI than that produced by RF, but the overall accuracy was incriminated by poor classification of SA. Therefore, CNN obtained slightly higher OA than that of RF. Some differences were observed between CNN classified maps when combined data of LiDAR with two WV-3 images and LiDAR with HSI were input. Results from combined data (Fig. 12 p and q) depicted the distribution of KO and AI better than using WV-3 images alone (Fig. 12 i and j), but it performed relatively poorer when using LiDAR data alone (Fig. 12 l). While CNN produced similar mapping results compared to RF when applying combined LiDAR and HSI (Fig. 12 o and r). In contrast to RF, CNN tended to rely on optical images more than LiDAR grid layers when multi-datasets were used. Overall, the four maps derived from the WV-3 MS + LiDAR dataset with RF (Fig. 12 m), the WV-3 PS + LiDAR dataset with RF (Fig. 12 n), and HSI + LiDAR dataset with both RF and CNN (Fig. 12 o and r) provided the most satisfactory results in this experiment.

5. Discussion

5.1. Vertical stratification

The proposed vertical stratification algorithm based on the *CHM* objects separated understory and overstory LiDAR returns effectively. Previous studies binned returns into horizontal grids or individual canopy to plot height histogram (Duncanson et al., 2014; Hamraz et al., 2017). We attempted to segment individual trees in initial experiments but the result was inaccurate because the tree crowns were relatively flat in the mangrove habitat. Since the shrub layers were distinctively lower than the arbors in multi-layered areas, the *CHM* segmentation favorably demarcated regions of single-layered and multi-layered.

Even though our vertical stratification result showed a better correlation between estimated and measured understory height comparing to the previous study (Hamraz et al., 2017), some constraints may lead to discrepancies in the estimation. Heights of understory under AM were frequently underestimated due to AM defoliation in summer from May to July in 2018. LiDAR data was acquired before the defoliation in March 2018 while canopy height measurement was conducted after defoliation in January 2019. The understory of AM was observed to grow rapidly during the defoliation period with sufficient sunlight. Another potential constraint might be resulted from the mixed overstory and understory returns. Areas with a height threshold larger than 2.75 m might misclassify some returns from branches or trunks of tall trees as understory returns, which led to the overestimation of understory canopy height.

In the past, some scholars have investigated the stratified profile of mangroves through intensive field measurements. Even though the spatial coverage is limited, Duke and Khan (1993) described the mangrove structure and composition of a transect across inland to seaward fringe in Mai Po, which are consistent with our vertical stratification result. Our proposed procedure applied the LiDAR data first to provide an overview of the vertical structure of both overstory and understory. It is the first document that captures both the species distribution and canopy height profile of the undergrowth in Mai Po core area. Notably, it also indicates the area of single-layered canopies without undergrowth, where they are usually found in the seaward fringe with limited understanding about their vertical structure and species in the past. With improved understanding of understory distribution, it is able to quantify the mangrove habitats with more accurate estimation of biophysical parameters such as leaf area index and biomass, which mainly relied on overstory metrics in the past.

5.2. Species classification

The result of feature selection suggested the WV-3 visible bands from green to red edge region were useful to discriminate some mangrove

species. The broad spectral region is possible to collect more spectral signals under the canopy surface comparing to the narrow spectral region. Broadband data is more sensitive to the spectral properties of the background but less sensitive to canopy architecture (Broge and Leblanc, 2001), which is conducive to identify those species with high variation in canopy architectures. As an example in this study, AM often had thin foliage and bent branches resultant in high variation in spectral properties in individual crowns. Mixed signals from understories, branches, and canopy reduced the spectral heterogeneity of AM, thus, produced smooth patches and improved the identification accuracy. However, the separation of KO and KOAI mainly relied on the red edge and NIR bands of WV-3 data. The multi-layer canopy structure would cause stronger additive scattering effect (Eriksson et al., 2006; Treitz et al., 2010) and results in higher reflectance. But the confusion between KO and KOAI was still observed when using optical data alone, which signified that optical data can only fairly indicate the presence of understory in our case.

Red edge bands located at different wavelength ranges are sensitive to different species (Treitz et al., 2010). Our analysis of HSI data also found that subtle differences among the mangrove species were observed in red edge bands with representative wavelengths at 704 nm, 744 nm, 695 nm, and 700 nm. Various red edge bands were selected that might echo the proposed hypothesis as once sufficient data exists for species within a specific region, which indicates the red edge region in our case, these wavelengths can be used to assist the discrimination between similar vegetation species directly and statistically (Cochrane, 2000). Similar to previous studies (Dalponte et al., 2009), the significant role of red edge bands in species classification was also revealed when using WV-3 images and HSI.

This study sought to compare the performance of three classifiers RF, SVM, and CNN with various data combinations. RF and SVM obtained relatively low overall accuracy 0.63–0.70 with sole input of optical images. Similar to the previous study (Hartling et al., 2019), our results suggested that CNN yielded higher accuracy than RF and SVM in mangrove species classification and produced a higher OA of 0.82 when fine resolution optical images of WV-3 PS and HSI were used. The high spatial resolution could become a disadvantage for traditional pixel-based classification because forest characteristics such as multi-layered structure and small tree crown might lead to mixed pixels and result in misclassification (Immitzer et al., 2012; Kamal and Phinn, 2011). On the other hand, CNN applies the convolution calculation, which considers surrounding spatial contexts for prediction. High-resolution can offer rich spatial and textural detail to conform convolution calculation, which is able to improve mapping accuracy. In terms of classification disagreement, CNN generated lower AD than RF and SVM, which could be related to the fact that the classified map was smoothed by reducing isolated pixels, which was always pronounced in pixel-based classification. However, CNN yielded a similar overall accuracy of 0.80 compared to RF (OA) = 0.78 and SVM (OA) = 0.79 when using LiDAR metrics. The declined accuracy of CNN was mainly due to errors in AI and grass estimation and overestimated AC and SA when LiDAR metrics were input. Shrub AI and grass with small crown size were often congregated to heterogeneous and narrow areas. The 2 m LiDAR metrics were not fine enough to provide detailed spatial context for CNN to identify AI and grass accurately. AC and SA sporadically appeared within clusters of other species around the riverside. With input patches of 2 m resolution, other species or water areas might interfere the prediction. Meanwhile, the smoothing mechanism of CNN tended to expand the misclassified pixels in the surrounding area. Apart from the resolution of data, the quantity of training samples could impact the classification result. Since deep learning algorithms typically required a large sample set to build an effective model (Hartling et al., 2019; Maxwell et al., 2018), the limited samples for the three targeted species including AC, SA, and grass might hinder model computation. Especially for SA as no testing sample were identified correctly by the CNN classifier. Some considerations can also be made when CNN

method is used such as parameters fine-tuning and CNN structure modification. Moreover, applying CNN directly to LiDAR returns instead of metrics layers would be a potential approach. In terms of computational efficiency, RF and SVM were more efficient. Although they required some time for parameter turning during the stage of model training, but once trained, they can efficiently map the entire area since they were robust to parameter setting. While CNN was implemented based on patches which required additional computation time and relatively powerful processing hardware.

6. Conclusions

This study conducted mangrove species classification based on stratifying the overstory and understory canopy in a significant mangrove habitat within the territory of Hong Kong. The result demonstrated that LiDAR data and the proposed stratification approach can effectively stratify the overstory and understory of mangrove stands. This study produced the first map of the understory in this ecologically important and sensitive habitat. It was found that knowledge of understory distribution and height profile provided valuable information for species classification. Combined canopy information of multi-layers would also be useful to improve the estimation of biophysical parameters such as leaf area index and biomass, which has long been used for biodiversity and wildlife habitat management.

High-resolution multispectral image WV-3, airborne HSI, LiDAR data, and the combination of them were applied in three classifiers RF, SVM, and CNN. The feature selection process not only improved classification efficiency and accuracy but also helped identify important and salient features that were essential for species differentiation. Spectral and LiDAR features had their respective advantages and drawbacks for species discrimination. As shown from the results, spectral features from optical images and structural features extracted from LiDAR metrics were complementary to each other, which was able to improve the classification accuracy. Classification results from RF and SVM stressed the important role of LiDAR features while CNN tended to rely on spectral features when multi-data were used. Satisfactory mapping results (OA = 0.84–0.88) were produced using either WV-3 and LiDAR data with RF and SVM or HSI and LiDAR data with all three classifiers in this study. The synergy of LiDAR with WV-3 or with HSI produced comparable mapping results, which indicated both HSI and WV-3 images are suitable for large scale species identification and mapping while the potential of HSI can be further explored.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rse.2021.112403>.

References

- Abdel-Rahman, E.M., Mutanga, O., Adam, E., Ismail, R., 2014. Detecting Sirex noctilio grey-attacked and lightning-struck pine trees using airborne hyperspectral data, random forest and support vector machines classifiers. *ISPRS J. Photogramm. Remote Sens.* 88, 48–59. <https://doi.org/10.1016/j.isprsjprs.2013.11.013>.
- AFCED, 2019. Agriculture, Fisheries and Conservation Department [WWW Document]. URL: https://www.afcd.gov.hk/english/conservation/con_wet/con_wet_look/con_wet_look_gen/con_wet_look_gen.html (accessed 5.29.19).
- Axelsson, P., 2000. DEM Generation from Laser Scanner Data Using Adaptive TIN Models. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* 110–117.
- Ballanti, L., Blesius, L., Hines, E., Kruse, B., 2016. Tree Species Classification Using Hyperspectral Imagery: A Comparison of Two Classifiers. *Remote Sens.* 8, 445. <https://doi.org/10.3390/rs8060445>.
- Blackburn, G.A., 1998. Quantifying chlorophylls and carotenoids at leaf and canopy scales: an evaluation of some hyperspectral approaches. *Remote Sens. Environ.* 66, 273–285. [https://doi.org/10.1016/S0034-4257\(98\)00059-5](https://doi.org/10.1016/S0034-4257(98)00059-5).
- Blasco, F., Bellan, M.F., Chaudhury, M.U., 1992. Estimating the extent of floods in Bangladesh using SPOT data. *Remote Sensing of Environment* 39 (3), 167–178. [https://doi.org/10.1016/0034-4257\(92\)90083-V](https://doi.org/10.1016/0034-4257(92)90083-V).
- Breiman, L., 2001. Random Forests. *Mach. Learn.* 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.
- Broge, N.H., Leblanc, E., 2001. Comparing prediction power and stability of broadband and hyperspectral vegetation indices for estimation of green leaf area index and canopy chlorophyll density. *Remote Sens. Environ.* 76, 156–172. [https://doi.org/10.1016/S0034-4257\(00\)00197-8](https://doi.org/10.1016/S0034-4257(00)00197-8).
- Burai, P., Deák, B., Valkó, O., Tomor, T., 2015. Classification of Herbaceous Vegetation Using Airborne Hyperspectral Imagery. *Remote Sens.* 7, 2046–2066. <https://doi.org/10.3390/rs70202046>.
- Chadwick, J., 2011. Integrated LiDAR and IKONOS multispectral imagery for mapping mangrove distribution and physical properties. *Int. J. Remote Sens.* 32, 6765–6781. <https://doi.org/10.1080/01431161.2010.512944>.
- Chan, P.K., Ching, S.H., Erfemeijer, P.L.A., 2011. Developing a mangrove management strategy in the estuaries of Deep Bay, Shan Pui River and Tin Shui Wai Drainage. *Channel* 1–15.
- Cheng, Q., Varshney, P.K., Arora, M.K., 2006. Logistic regression for feature selection and soft classification of remote sensing data. *IEEE Geosci. Remote Sens. Lett.* 3, 491–494. <https://doi.org/10.1109/LGRS.2006.8779949>.
- Chun, B.B., Keat, S.C., Syahreza, S., Jafri, M.Z.M., San, L.H., 2015. Discrimination of mangrove species in Matang Mangrove Forest Reserve, Perak using in-situ measurement of hyperspectral leaf reflectance. *AIP Publishing* 110004. <https://doi.org/10.1063/1.4915223>.
- Chung, K., Territories, N., 2013. Mai Po nature reserve habitat management. In: *Monitoring and Research Plan 2013–2018, Vol. 1.1*.
- Clevers, J.G.P.W., 1999. The use of imaging spectrometry for agricultural applications. *ISPRS J. Photogramm. Remote Sens.* 54, 299–304. [https://doi.org/10.1016/S0924-2716\(99\)00033-7](https://doi.org/10.1016/S0924-2716(99)00033-7).
- Cochrane, M.A., 2000. Using vegetation reflectance variability for species level classification of hyperspectral data. *Int. J. Remote Sens.* 21, 2075–2087. <https://doi.org/10.1080/01431160050021303>.
- Dalponte, M., Bruzzone, L., Vescovo, L., Gianelle, D., 2009. The role of spectral resolution and classifier complexity in the analysis of hyperspectral images of forest areas. *Remote Sens. Environ.* 113, 2345–2355. <https://doi.org/10.1016/j.rse.2009.06.013>.
- Duke, N.C., Khan, M.A., 1993. Structure and composition of the seaward mangrove forest at Mai Po Marshes Nature Reserve, Hong Kong. *Proc. Int. Work. Mangrove Ecosyst. Deep Bay Mai Po Marshes*, pp. 83–104.
- Duncanson, L.I.L., Cook, B.D.D., Hurr, G.C., Dubayah, R.O.O., 2014. An efficient, multi-layered crown delineation algorithm for mapping individual tree structure across multiple ecosystems. *Remote Sens. Environ.* 154, 378–386. <https://doi.org/10.1016/j.rse.2013.07.044>.
- Eriksson, H.M., Eklundh, L., Kuusk, A., Nilson, T., 2006. Impact of understory vegetation on forest canopy reflectance and remotely sensed LAI estimates. *Remote Sens. Environ.* 103, 408–418. <https://doi.org/10.1016/j.rse.2006.04.005>.
- Fan, R.-E., Chen, P.-H., Lin, C.-J., 2005. Working Set Selection Using Second Order Information for Training Support Vector Machines. *J. Mach. Learn. Res.* 6, 1889–1918.
- FAO, 2007. The World's Mangroves 1980-2005. *FAO For. Pap.* 153, p. 89 <https://doi.org/978-92-5-105856-5>.
- Fassnacht, F.E., Neumann, C., Forster, M., Buddenbaum, H., Ghosh, A., Clasen, A., Joshi, P.K., Koch, B., 2014. Comparison of feature reduction algorithms for classifying tree species with hyperspectral data on three central European test sites. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 7, 2547–2561. <https://doi.org/10.1109/JSTARS.2014.2329390>.
- Féret, J.-B.B., Gitelson, A.A.A., Noble, S.D.D., Jacquemoud, S., 2017. PROSPECT-D: towards modeling leaf optical properties through a complete lifecycle. *Remote Sens. Environ.* 193, 204–215. <https://doi.org/10.1016/j.rse.2017.03.004>.
- Fricker, G.A., Ventura, J.D., Wolf, J.A., North, M.P., Davis, F.W., Franklin, J., 2019. A convolutional neural network classifier identifies tree species in mixed-conifer forest from hyperspectral imagery. *Remote Sens.* 11, 2326. <https://doi.org/10.3390/rs11192326>.
- Ghiyam, A., Shafri, H.Z.M., 2010. A review on hyperspectral remote sensing for homogeneous and heterogeneous forest biodiversity assessment. *Int. J. Remote Sens.* <https://doi.org/10.1080/01431160902926681>.
- Ghosh, A., Joshi, P.K.K., 2014. A comparison of selected classification algorithms for mapping bamboo patches in lower Gangetic plains using very high resolution

- WorldView 2 imagery. *Int. J. Appl. Earth Obs. Geoinf.* 26, 298–311. <https://doi.org/10.1016/j.jag.2013.08.011>.
- Giri, S., Mukhopadhyay, A., Hazra, S., Mukherjee, S., Roy, D., Ghosh, S., Ghosh, T., Mitra, D., 2014. A study on abundance and distribution of mangrove species in Indian Sundarban using remote sensing technique. *J. Coast. Conserv.* 18, 359–367. <https://doi.org/10.1007/s11852-014-0322-3>.
- Green, E.P., Clark, C.D., Mumby, P.J., Edwards, A. J., Ellis, A. C., 1998. Remote sensing techniques for mangrove mapping. *Int. J. Remote Sens.* 19, 935–956. <https://doi.org/10.1080/014311698215801>.
- Guyon, I., Elisseeff, A., 2003. An introduction to variable and feature selection. *J. Mach. Learn. Res.* 3, 1157–1182.
- Hamraz, H., Contreras, M.A., Zhang, J., 2017. Vertical stratification of forest canopy for segmentation of understory trees within small-footprint airborne LiDAR point clouds. *ISPRS J. Photogramm. Remote Sens.* 130, 385–392. <https://doi.org/10.1016/j.isprsjprs.2017.07.001>.
- Hartling, S., Sagan, V., Sidike, P., Maimaitjiang, M., Carron, J., Hartling, S., Sagan, V., Sidike, P., Maimaitjiang, M., Carron, J., 2019. Urban tree species classification using a worldview-2/3 and LiDAR data fusion approach and deep learning. *Sensors (Switzerland)* 19, 1–23. <https://doi.org/10.3390/s19061284>.
- Hughes, G.F., 1968. On the mean accuracy of statistical pattern recognizers. *IEEE Trans. Inf. Theory* 14, 55–63. <https://doi.org/10.1109/TTT.1968.1054102>.
- Immitzer, M., Atzberger, C., Koukal, T., 2012. Tree species classification with random Forest using very high spatial resolution 8-band WorldView-2 satellite data. *Remote Sens.* 4, 2661–2693. <https://doi.org/10.3390/rs4092661>.
- Jones, T.G., Coops, N.C., Sharma, T., 2010. Assessing the utility of airborne hyperspectral and LiDAR data for species distribution mapping in the coastal Pacific Northwest, Canada. *Remote Sens. Environ.* 114, 2841–2852. <https://doi.org/10.1016/j.rse.2010.07.002>.
- Ju, J., Kolaczky, E.D., Gopal, S., 2003. Gaussian mixture discriminant analysis and sub-pixel land cover characterization in remote sensing. *Remote Sens. Environ.* 84, 550–560. [https://doi.org/10.1016/S0034-4257\(02\)00172-4](https://doi.org/10.1016/S0034-4257(02)00172-4).
- Kamal, M., Phinn, S., 2011. Hyperspectral data for mangrove species mapping: a comparison of pixel-based and object-based approach. *Remote Sens.* 3, 2222–2242. <https://doi.org/10.3390/rs3102222>.
- Khan, M.N.I., Kabir, M.E., 2017. Ecology of *Kandelia obovata* (S., L.) Yong: A Fast-Growing Mangrove in Okinawa, Japan, pp. 287–301. https://doi.org/10.1007/978-4-431-56481-2_18.
- Khan, M.N.I., Sharma, S., Berger, U., Koedam, N., Dahdouh-Guebas, F., Hagihara, A., 2013. How do tree competition and stand dynamics lead to spatial patterns in monospecific mangroves? *Biogeosciences* 10, 2803–2814. <https://doi.org/10.5194/bg-10-2803-2013>.
- Kingma, D.P., Ba, J.L., 2015. Adam: A method for stochastic optimization, in: 3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings. International Conference on Learning Representations, ICLR.
- Li, X., Qi, W., 2015. An effective pansharpening method for WorldView-2 satellite images. In: Proceedings of 2015 International Conference on Estimation, Detection and Information Fusion, ICEDIF 2015. Institute of Electrical and Electronics Engineers Inc., pp. 88–92. <https://doi.org/10.1109/ICEDIF.2015.7280168>.
- Koenig, K., Höfle, B., 2016. Full-Waveform Airborne Laser Scanning in Vegetation Studies—A Review of Point Cloud and Waveform Features for Tree Species Classification. *Forests* 7, 198. <https://doi.org/10.3390/f7090198>.
- Li, H., Jing, L., Tang, Y., 2017. Assessment of Pansharpening methods applied to WorldView-2 imagery fusion. *Sensors* 17, 89. <https://doi.org/10.3390/s17010089>.
- Li, Q., Wong, F.K.K., Fung, T., 2019. Classification of mangrove species using combined WorldView-3 and LiDAR data in Mai Po Nature Reserve, Hong Kong. *Remote Sens.* 11, 1–17. <https://doi.org/10.3390/rs11182114>.
- Li, Q., Wong, F.K.K., Fung, T., 2019. Comparison Feature Selection Methods for Subtropical Vegetation Classification with Hyperspectral Data. *Int. Geosci. Remote Sens. Symp.* 3693–3696. <https://doi.org/10.1109/IGARSS.2019.8898541>.
- Liu, L., Coops, N.C., Aven, N.W., Pang, Y., 2017. Mapping urban tree species using integrated airborne hyperspectral and LiDAR remote sensing data. *Remote Sens. Environ.* 200, 170–182. <https://doi.org/10.1016/j.rse.2017.08.010>.
- Martinuzzi, S., Vierling, L.A., Gould, W.A., Falkowski, M.J., Evans, J.S., Hudak, A.T., Vierling, K.T., 2009. Mapping snags and understory shrubs for a LiDAR-based assessment of wildlife habitat suitability. *Remote Sens. Environ.* 113, 2533–2546. <https://doi.org/10.1016/j.rse.2009.07.002>.
- Maxwell, A.E., Warner, T.A., Fang, F., 2018. Implementation of machine-learning classification in remote sensing: an applied review. *Int. J. Remote Sens.* 39, 2784–2817. <https://doi.org/10.1080/01431161.2018.1433343>.
- Mutanga, O., Skidmore, A.K., 2007. Red edge shift and biochemical content in grass canopies. *ISPRS J. Photogramm. Remote Sens.* 62, 34–42. <https://doi.org/10.1016/j.isprsjprs.2007.02.001>.
- Peerbhay, K.Y., Mutanga, O., Ismail, R., 2013. Commercial tree species discrimination using airborne AISA Eagle hyperspectral imagery and partial least squares discriminant analysis (PLS-DA) in KwaZulu-Natal, South Africa. *ISPRS J. Photogramm. Remote Sens.* 79, 19–28. <https://doi.org/10.1016/j.isprsjprs.2013.01.013>.
- Pontius, R.G., Millones, M., 2011. Death to kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment. *Int. J. Remote Sens.* 32, 4407–4429. <https://doi.org/10.1080/01431161.2011.552923>.
- Raczko, E., Zagajewski, B., 2017. Comparison of support vector machine, random forest and neural network classifiers for tree species classification on airborne hyperspectral APEX images. *Eur. J. Remote Sens.* 50, 144–154. <https://doi.org/10.1080/22797254.2017.1299557>.
- Rapinel, S., Clément, B., Magnanon, S., Sellin, V., Hubert-Moy, L., 2014. Identification and mapping of natural vegetation on a coastal site using a Worldview-2 satellite image. *J. Environ. Manag.* 144, 236–246. <https://doi.org/10.1016/j.jenvman.2014.05.027>.
- Richter, R., Reu, B., Wirth, C., Doktor, D., Vohland, M., 2016. The use of airborne hyperspectral data for tree species classification in a species-rich Central European forest area. *Int. J. Appl. Earth Obs. Geoinf.* 52, 464–474.
- Shen, X., Cao, L., 2017. Tree-Species Classification in Subtropical Forests Using Airborne Hyperspectral and LiDAR Data. *Remote Sens.* 9, 1180. <https://doi.org/10.3390/rs9111180>.
- Shi, Y., Wang, T., Skidmore, A.K., Heurich, M., 2018. Important LiDAR metrics for discriminating forest tree species in Central Europe. *ISPRS J. Photogramm. Remote Sens.* 137, 163–174. <https://doi.org/10.1016/j.isprsjprs.2018.02.002>.
- Singh, K.K., Davis, A.J., Meentemeyer, R.K., 2015. Detecting understory plant invasion in urban forests using LiDAR. *Int. J. Appl. Earth Obs. Geoinf.* 38, 267–279. <https://doi.org/10.1016/J.JAG.2015.01.012>.
- Sothe, C., de Almeida, C.M., Schimanski, M.B., Liesenberg, V., 2018. Integration of Worldview-2 and lidar data to MAP a subtropical forest area: comparison of machine learning algorithms. In: IGARSS 2018–2018 IEEE International Geoscience and Remote Sensing Symposium. IEEE, pp. 6207–6210. <https://doi.org/10.1109/IGARSS.2018.8517941>.
- Stagakis, S., Markos, N., Sykioti, O., Kyparissis, A., 2014. Tracking seasonal changes of leaf and canopy light use efficiency in a *Phlomis fruticosa* Mediterranean ecosystem using field measurements and multi-angular satellite hyperspectral imagery. *ISPRS J. Photogramm. Remote Sens.* 97, 138–151. <https://doi.org/10.1016/j.isprsjprs.2014.08.012>.
- Sun, Z., Zhao, X., Wu, M., Wang, C., 2019. Extracting urban impervious surface from WorldView-2 and airborne LiDAR data using 3D convolutional neural networks. *J. Indian Soc. Remote Sens.* 47, 401–412. <https://doi.org/10.1007/s12524-018-0917-5>.
- Tam, N.F.Y., Wong, Y.-S., 2000. Hong Kong mangroves. City University of Hong Kong Press.
- Teobaldelli, M., Cona, F., Saulino, L., Migliozi, A., D'Urso, G., Langella, G., Manna, P., Saracino, A., 2017. Detection of diversity and stand parameters in Mediterranean forests using leaf-off discrete return LiDAR data. *Remote Sens. Environ.* 192, 126–138. <https://doi.org/10.1016/j.rse.2017.02.008>.
- Treitz, P., Thomas, V., Zarco-Tejada, P.J., Gong, P., Curran, Paul J., 2010. Hyperspectral remote sensing for forestry. *ASPRS*.
- Wang, L., Sousa, W.P., Gong, P., 2004a. Integration of object-based and pixel-based classification for mapping mangroves with IKONOS imagery. *Int. J. Remote Sens.* 25, 5655–5668. <https://doi.org/10.1080/014311602331291215>.
- Wang, L., Sousa, W.P., Gong, P., Biging, G.S., 2004b. Comparison of IKONOS and QuickBird images for mapping mangrove species on the Caribbean coast of Panama. *Remote Sens. Environ.* 91, 432–440. <https://doi.org/10.1016/j.rse.2004.04.005>.
- Wang, T., Zhang, H., Lin, H., Fang, C., 2015. Textural-spectral feature-based species classification of mangroves in Mai Po nature reserve from Worldview-3 imagery. *Remote Sens.* 8, 24. <https://doi.org/10.3390/rs8010024>.
- Waser, L.T., Küchler, M., Jütte, K., Stampfer, T., Waser, L.T., Küchler, M., Jütte, K., Stampfer, T., 2014. Evaluating the potential of WorldView-2 data to classify tree species and different levels of ash mortality. *Remote Sens.* 6, 4515–4545. <https://doi.org/10.3390/rs6054515>.
- Wing, B.M., Ritchie, M.W., Boston, K., Cohen, W.B., Gitelman, A., Olsen, M.J., 2012. Prediction of understory vegetation cover with airborne lidar in an interior ponderosa pine forest. *Remote Sens. Environ.* 124, 730–741. <https://doi.org/10.1016/J.RSE.2012.06.024>.
- Wong, F.K.K., Fung, T., 2014. Combining EO-1 Hyperion and Envisat ASAR data for mangrove species classification in Mai Po Ramsar site, Hong Kong. *Int. J. Remote Sens.* 35, 7828–7856. <https://doi.org/10.1080/01431161.2014.978034>.
- Zhang, Z., Liu, X., 2013. WorldView-2 Satellite Imagery and Airborne LiDAR Data for Object-Based Forest Species Classification in a Cool Temperate Rainforest Environment. Springer, Berlin, Heidelberg, pp. 103–122. https://doi.org/10.1007/978-3-642-36379-5_7.
- Zhao, K., Popescu, S., 2009. Lidar-based mapping of leaf area index and its use for validating GLOBCARBON satellite LAI product in a temperate forest of the southern USA. *Remote Sens. Environ.* 113, 1628–1645. <https://doi.org/10.1016/J.RSE.2009.03.006>.
- Zhou, W., Huang, G., Troy, A., Cadenasso, M.L., 2009. Object-based land cover classification of shaded areas in high spatial resolution imagery of urban areas: a comparison study. *Remote Sens. Environ.* 113, 1769–1777. <https://doi.org/10.1016/j.rse.2009.04.007>.
- Raczko, E., Zagajewski, B., 2017. Comparison of support vector machine, random forest and neural network classifiers for tree species classification on airborne hyperspectral APEX images. *Eur. J. Remote Sens.* 50, 144–154. <https://doi.org/10.1080/22797254.2017.1299557>.